# Ensemble-Based Information Retrieval With Mass Estimation for Hyperspectral Target Detection

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***Abstract*— Given the prior information of the target, hyperspectral target detection focuses on exploiting spectral differences to separate objects of interest from the background, which can be treated as information retrieval (IR) task in machine learning (ML). Most traditional detection methods work in the original feature space and rely heavily on specific assumptions, which cannot guarantee effective extraction of features for the target and background in hyperspectral images (HSIs). Mass estimation (ME) is a base modeling mechanism that has been proven to effectively solve problems in IR and is not restricted by specific assumptions. In this article, we propose a novel target detection method through ensemble-based IR with ME (EIRME). By directly deriving the ordering from a sample set to rank data points, ME provides a simple and straightforward ranking measure to ensure that points similar to the given target are far away from dissimilar points. For the estimation of mass distribution, the proposed method utilizes a tree-structured mapping to generate a feature space, in which the separability of the target and background is further improved. In particular, to break through the technical difficulty that the direct migration of IR methods with mass measure cannot specifically meet the high-precision requirements of target detection in HSIs, we develop a specialized measurement, topological mass, which innovatively combines the mass measure with tree topology to quantify the spectral difference for detection output. Moreover, the IR with ME based on parallel measurements through ensemble trees provides a robust solution with better generalization capacity and higher precision for hyperspectral target detection, facilitating practical applications. Experimental results on benchmark HSI datasets prove that the specialized measurement that we developed successfully overcomes the drawbacks of the direct migration of IR methods with ME and exhibits unique advantages. In addition, comparisons with the most classic and advanced detection algorithms demonstrate the superiority of the proposed method.**

***Index Terms*— Hyperspectral image (HSI), information retrieval (IR), machine learning (ML), mass estimation (ME), target detection.**

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## I. Introduction

HYPERSPECTRAL sensors capture specific scenes on the continuous and subdivided electromagnetic spectrum (0.4–2.5 $\mu$m) spanning the range of visible light, near-infrared, and short-wave infrared [1], [2]. Hyperspectral images (HSIs) provide both spatial and spectral information for multiple types of land cover, which significantly improves the quality, detail, reliability, and credibility of data analysis, making it highly valuable for academic research and practical applications [3], [4]. As one of the most active topics in hyperspectral remote sensing, target detection uses abundant spectral information to locate objects of interest in specific scenes [5]–[7]. So far, hyperspectral target detection has been widely used and exerted its unique advantages in a variety of military and civilian fields, such as exposing camouflage, identifying small targets, and determining the category of land cover [8]–[10]. For practical considerations, it is necessary to develop target detection methods suitable for HSIs to meet the performance requirements of low false alarm rate (FAR), high detection rate, and robustness [11], [12]. In the past few decades, a large number of detection algorithms migrated from signal processing and machine learning (ML) have emerged in this field [6], [13], [14]. The rapid development of hyperspectral target detection faces various challenges and encourages the emergence of different problem-solving ideas and strategies [12], [15], [16].

With the availability of the prior information of the target, the essential problem of hyperspectral target detection is to separate the target of interest from the background based on the spectral difference between different types of land covers [1]. More specifically, the design of the detector attempts to find a scientific and effective way to measure and quantify the spectral difference for detection output in order to highlight the target and suppress the background. Detection methods based on signal processing have developed rapidly since the 1990s and laid a solid foundation for this discipline [1]. The general design process of such traditional methods is to derive a decision function through specific model assumptions

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and decision criteria and then to substitute test pixels into the function to obtain the detection output for determining the existence of the target [15], [17]. The probability statistics (whited-space) model and subspace projection model are the most classic and important model assumptions in the field of signal processing [18],–[20]. Algorithms developed based on such models have shown stable detection effects in many cases. Constrained energy minimization (CEM) is one of the most widely used detectors derived from a linearly constrained minimum-variance beamformer in digital signal processing [21]. It extracts target signals in a specific direction by suppressing background signals in other directions using the Fisher linear discriminant criterion [22]. The generalized likelihood ratio test (GLRT) depends on the assumptions of the conditional probability distribution for the target and the background [23]. Its detection performance has the properties of high signal-to-noise ratio (SNR) and constant FAR (CFAR) [20], [24], which makes the algorithm more robust in the use of HSIs in various real scenes [25]. There are two adaptive versions of GLRT [26]: the adaptive coherence/cosine estimator (ACE) and the adaptive matched filter (AMF). Their decision functions are derived using the correlation or covariance matrix of the given HSI [21], [27]. The orthogonal subspace projection (OSP) and its derivative algorithms use mathematical projection operators to project the test pixels into the orthogonal subspace of the background to highlight the target and eliminate interference [28], [29]. In [30], a seminal joint and progressive subspace analysis (JPSA) framework is proposed to extract dimension-reduced HSI features from spatially and spectrally aligned manifolds, achieving state-of-the-art results in subsequent applications. Tao *et al.* [31] proposed a novel detection method using fractional Fourier transform to obtain discriminant features between targets and background, which also provided excellent detection effect. It is worth noting that such detectors can only exhibit ideal performance when the background spectra are accurately acquired. The detectors would suffer performance degradation under the condition of limited prior information. Based on signal processing, HSI target detection has established a mature theoretical system and various algorithms [1], [6]. However, the traditional methods rely heavily on specific assumptions and have limited analytical capabilities for complex hypothetical models when performing feature extraction on HSIs containing multiple land cover classes. Therefore, they cannot make full use of the abundant detailed information provided by high-spectral-resolution images, resulting in poor target detection effects [32]–[34].

In the past few years, the rapid development of ML has inspired a new surge of activities in hyperspectral target detection, and the corresponding new algorithms have exerted their unique advantages in many cases [35], [36]. It is quite challenging to learn from and apply ML on the basis of existing theoretical systems to adapt to target detection for HSIs in various highly complex scenes [6], [37]–[40]. So far, the research hotspots for method migration in ML include kernel-based detectors for nonlinear signal processing [41], [42], representation-based detectors according to the compressed sensing theory [43]–[46], deep learning methods based on artificial neural networks [36], [47], and so on. In particular, recently, based on the powerful ability to learn deep features of high-dimensional data [48]–[50], deep learning has played an unparalleled advantage in data mining and information extraction tasks [51]–[54]. Various neural networks, such as a convolutional neural network (CNN) [53], [55], an autoencoder (AE) [56], [57], a generative adversarial network (GAN) [58], [59], a deep belief network (DBN) [60], [61], and their derivative versions, have been increasingly introduced into HSI processing and analysis [62]–[64]. Li *et al.* [36] first applied deep CNN to anomaly detection in HSIs and proposed a novel anomaly detection framework with transferred deep CNN. After that, a tremendous amount of research on data-driven training networks has gradually revealed the great potential of deep learning in the field of hyperspectral target detection [65]–[67]. All the aforementioned target detection methods focus on different application problems and satisfy different usage requirements, which also helps to inspire more thinking modes and problem-solving viewpoints in this field.

Information retrieval (IR) refers to the process in which users find out relevant information for a given query from a collection of information by means of retrieval tools according to application requirements. With the given prior knowledge of the target, hyperspectral target detection is dedicated to finding the location of the target in a specific scene. Regarding the HSI as a collection of information and the target spectrum as a given query, and finding the location of the target as a requirement, then target detection in the HSI is essentially an IR task [68]–[70]. Generally, original information collections, such as documents, pictures, videos, and audios, are converted into computer languages and stored in a database. After the user enters a query request, the retrieval system searches the database for information related to the query. Specifically, the similarities between all information and the query are quantified through a certain matching mechanism; then, the information is ranked in descending order and output after conversion. According to such principles of IR, the ranking of information is the basis for implementing this task, and a ranking measure that separates relevant information from irrelevant information is the key. With this idea, the core problem of designing a detector is to find an effective measure to ensure that points that are similar to the given target (i.e., the relevant information belonging to the target) are far away from dissimilar points (i.e., the irrelevant information belonging to the background). The ranking measure required by the IR task needs to fully utilize and quantify the spectral difference between the target and the background to achieve their separation [71].

Mass estimation (ME) as a base modeling mechanism has been utilized to solve various tasks in ML, such as IR, anomaly detection, and regression [68], [72]. Although target detection in HSI can be regarded as an IR task, the hyperspectral data contain abundant detailed information about land cover [73], [74] and involve rigid demands for the extraction of effective features of the target and background during processing. Correspondingly, the requirements for detection accuracy are very high. Therefore, there are few studies on applying ME

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to hyperspectral target detection. In this article, we propose a novel target detection method through ensemble-based IR with ME (EIRME). Specifically, ME uses a certain prescribed ordering derived from the data itself to rank the data cloud, which distinguishes fringe points from core points. For an HSI, the object of interest for target detection exists in the form of a few distinct data points, which corresponds to the fringe points. Conversely, the background corresponds to the core points. Based on this, in the case of a given query, that is, the given target spectrum, IR with ME can effectively separate the target from the background. Moreover, the binary tree is a powerful hierarchical structure that can describe mass distribution [75], [76], so a tree-structured mapping is exploited to ensure that the target and background are farther away from each other in the feature space. For the detection output, the spectral difference is quantified based on a combination of mass measure and tree topology. The specialized measurement that we developed is referred to as the topological mass. The proposed method is performed on ensemble trees to ensure robustness in view of the generalization error generated in the process of constructing a binary tree through random subsampling. We refer to the proposed hyperspectral target detection approach as EIRME.

At present, the research on target detection for HSIs is focused on the core technical difficulties, trying to find a simple, straightforward, and effective way to measure the spectral differences between different types of land cover to solve the problem of highlighting the target and suppressing the background. Such issues provide a wide space for the migration of methods from various fields. Unlike most traditional methods, the proposed EIRME is not limited to specific assumptions in the original feature space and manages to implement the detection based on ME instead of likelihood ratio or expensive distance calculations, such as the Mahalanobis distance or the Euclidean distance. The related work provides a solution to the technical problems of hyperspectral target detection from a new perspective. Specifically, the novelty and contributions of this article are summarized as follows:

1) To the best of our knowledge, EIRME is the first to apply IR with ME in the field of hyperspectral target detection. Without relying on traditional model assumptions but focusing on the features of high-dimensional data itself, hyperspectral target detection is regarded as an IR task. Herein, ME is exploited to solve this task. For an HSI, the properties of mass distribution ensure that the pixels from the target are stretched away from the background, and a simple and effective ranking measure provided by ME meets the essential requirements of the IR task.
2) For the effective extraction of features for the target and background in HSIs, a powerful hierarchical structure for high-dimensional data, the binary tree, is utilized to estimate the mass distribution. A tree-structured mapping is utilized to generate a feature space, or mass space, in which the separability of the target and the background is significantly improved. Each tree corresponds to a mass distribution estimated from the characteristics of the hyperspectral data; hence, the IR with ME based on an ensemble of trees can fit the characteristics of HSIs in various scenes for practical applications.
3) The direct migration of the IR method with mass measure cannot specifically meet the high-precision requirements for target detection in HSIs. A specialized measurement known as topological mass is developed to surpass this technical challenge. In the tree-structured feature space, the proposed method innovatively combines the mass measure with tree topology as a specialized ranking measure (i.e., topological mass) to take advantage of both. The developed measurement can effectively quantify spectral differences for detection output to further boost the detection accuracy.

The rest of this article is organized as follows. In Section II, a detailed description of the proposed EIRME is provided. Section III shows the experimental results on real hyperspectral data and gives performance analysis. Section IV draws the conclusions of this article.

## II. Ensemble-Based Information Retrieval With Mass Estimation

The processing flow of the proposed EIRME is shown in Fig. 1. The details of the methodology are elaborated in the following three parts: 1) ME implemented by mapping the HSI data to feature space and ranking data points for the IR task; 2) development of a specialized measurement for detection output by combining the mass measure with tree topology in the feature space; and 3) target detection as an IR task based on ensemble trees for parallel measurements. Sections II-A–II-C, respectively, explain these three parts in detail.

### *A. Multidimensional Mass Estimation for Information Retrieval*

ME is a base modeling mechanism that effectively solves data mining tasks, such as IR, anomaly detection, and regression [72]. It was initially designed to process 1-D data. Ting and Wells [77] extended the method to multidimensional data, making it more suitable for solving generic problems. At present, the research on ME has developed specific solutions for various tasks [68]. Given prior information of the target, target detection in HSIs can be regarded as an IR task with the query instance. Therefore, it is natural to think of employing ME as a powerful tool to overcome technical difficulties.

The core idea of ME is to stipulate an ordering for ranking points in a dataset according to the requirements of various tasks. Given the target spectrum, the objects of interest usually exist in the form of a few distinct pixels in a specific scene. The features of the target are completely different from that of the background in terms of spatial and spectral dimensions. Based on the modeling mechanism of mass, an HSI can be mathematically modeled as a data cloud, in which the core points with large mass values have a dense distribution and occupy a large proportion, while the fringe points with small mass values have a scattered distribution and occupy a small

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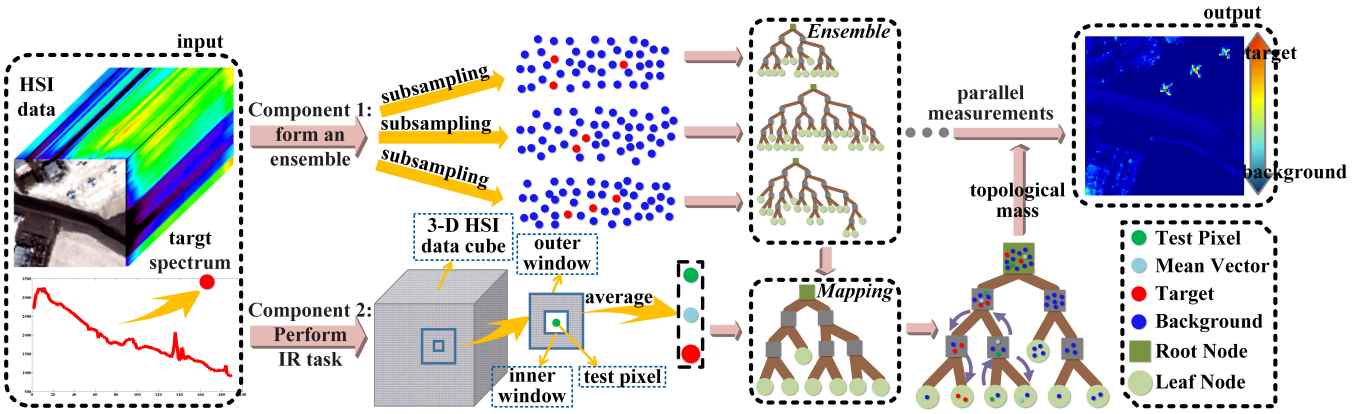


Fig. 1. Flowchart of the proposed EIRME for hyperspectral target detection.

proportion. Because of the differences in number, spatial distribution, and spectral characteristics, the target in the data cloud corresponds to fringe points, and the background corresponds to core points. By ranking data points in the HSI, ME strives to make points that are similar to the given query higher than those that are dissimilar, thereby keeping points from different types of the land cover away from each other for separation purposes. Similar points, which correspond to fringe points, are more likely to be judged as targets and are marked with smaller mass values. As for dissimilar points corresponding to the core points, they are more likely to be judged as backgrounds and marked with larger mass values. In general, ME provides a ranking measure for IR tasks to find interesting data points [78], [79]. Therefore, we can take advantage of ME to design algorithms for hyperspectral target detection that highlights the target and suppresses the background.

Data mass can be simply defined as the number of data points in a specific region, which is delimited by the mass distribution. The mass distribution is a concrete representation of an ordering derived directly from the sample set. The data mass from the core points to the fringe points can be estimated by generating a feature space, or mass space, in which there are multiple regions covering all points in a dataset [72], [77]. In the original feature space shown in Fig. 2(a), the data points with numerical separability are randomly distributed. The characteristics of a given query are embodied in the difference between its value and that of other data points in certain dimensions. When using a specific mapping to generate the feature space, the essence is to redistribute data through the division of regions and, ultimately, improve the separability between different types of points. Fig. 2(b) provides an intuitive display of a mass distribution, where different types of points are distinguished by size, shape, and color. Each rectangle in Fig. 2(c) represents a region, and different regions are set to different levels. The higher level regions have a smaller volume and mass values, while the lower level regions have larger volume and mass values. For a data point, the regions covering it overlap, which means that there are many points sharing multiple regions with it in the mass space.

Moreover, the mass value of the highest-level region shared with the given query can reflect the correlation between this

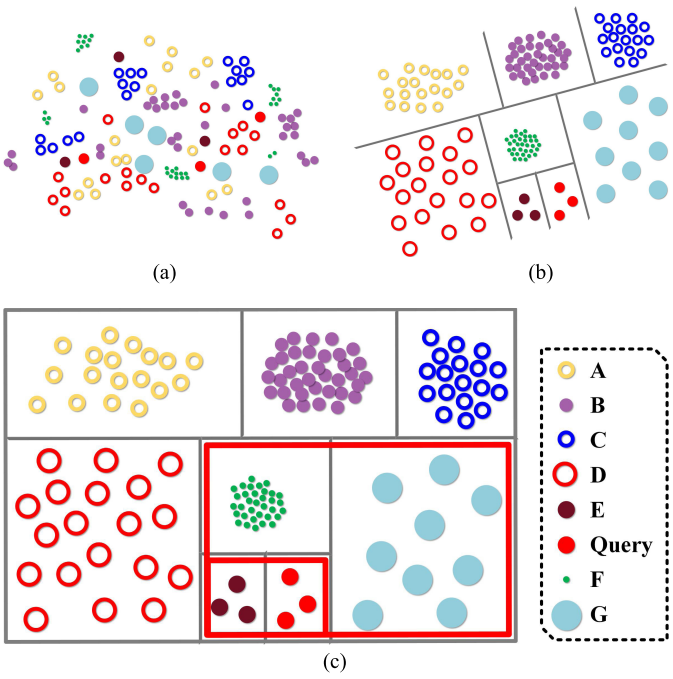


Fig. 2. (a) Data distribution in the original feature space. (b) Data distribution in the feature space. (c) Mass distribution.

point and the query. For example, the large and small red boxes in Fig. 2(c) represent the highest-level regions shared by the query with G and E, respectively, and they correspond to different mass values. It is obvious that the mass value of the region shared by the query and G is greater than that of E, which is more similar to the query. The smaller the mass value, the smaller the volume of the shared region, which indicates a stronger correlation. The above characteristics of mass distribution can be used to rank all points in a dataset to get their degree of relevance or similarity with a given query. This provides a feasible strategy for the implementation of IR.

The IR for multidimensional data corresponds to the target detection for HSIs. The execution procedure of ME for IR includes the following two components.

*1) Component 1:* A feature space is generated through a specific form of mapping to estimate the mass distribution.

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*2) Component 2:* The IR task is performed in the feature space or mass space.

For a given HSI, there is a huge spectral difference between the target and the background because they belong to different types of land cover, which is essentially reflected in the numerical difference. Based on this fact, the dataset can be mapped to a feature space, where the mass distribution greatly improves the separability of the target and the background. With the availability of the prior information of the target, IR is performed in the feature space, and pixels with high similarity to the given query can be found from an HSI containing a large number of pixels, which provides a reliable basis for the subsequent judgment of the existence of the target.

### *B. Development of Topological Mass*

The generation of a feature space determines the estimation of the mass distribution, which is critical for separating pixels from different types of land cover in hyperspectral data. How to stretch the fringe points corresponding to the target away from the core points corresponding to the background in the feature space through a specific mapping is essential to effectively highlight the target while suppressing the background. The binary tree is widely used as a powerful and practical hierarchical structure for processing multidimensional data in ML, which is perfect for estimating the mass distribution.

As mentioned earlier, data mass can be simply defined as the number of data points in a specific region. For a given HSI dataset, to perform an IR task with ME, the first is to use a specific form of mapping to generate a feature space to define specific regions. Based on the implementation of multidimensional ME described in [72], we adopted a tree-structured mapping in this article. From the mathematical perspective, the pixels in an HSI exist in the form of vectors, and the characteristics of a given target spectrum are embodied in which its values on some or all bands are larger or smaller than that of the background. Such characteristics of HSI data enable the separation of pixels from different types of land cover. The construction of the binary tree depends entirely on the characteristics of the data itself when using a tree-structured mapping to generate a feature space for separation. The process is carried out by randomly subsampling from the HSI, which is suitable for various scenes in practical applications. After completing the construction of a binary tree, the original dataset can be mapped to a feature space. Specifically, the original dataset is put into a root node, and the tree structure utilizes different levels of nodes to continuously divide the original dataset into disjoint subsets until it ends at leaf nodes. In the feature space, each node on the tree is a defined specific region, and the number of data points in this region is its mass value. Fig. 3 shows a constructed binary tree. For an HSI, put the entire dataset into the root node. Starting from the root node and ending with the leaf nodes, the tree structure utilizes different levels of nodes to continuously divide the original HSI dataset into disjoint subsets, and each division adds new regions into the feature space. This is an intuitive display of a simplified mass distribution estimated by the tree-structured mapping. In Fig. 3, each node on the tree

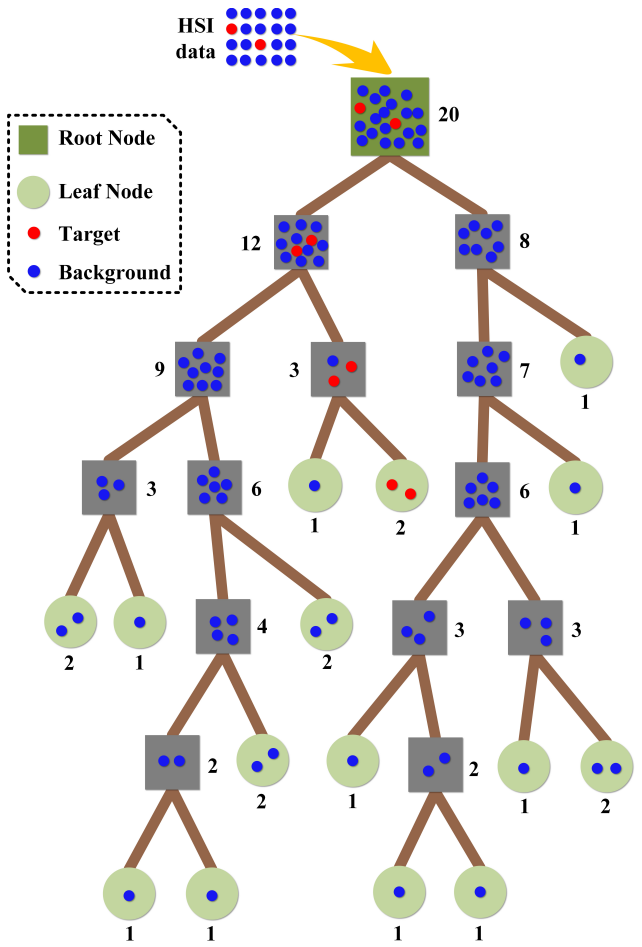


Fig. 3. Mass distribution estimated by the tree-structured mapping.

corresponds to a region, and the number of data points in this region provided next to it is its mass value. The higher the level of a region, or node, the smaller its mass value. The whole process gradually gathers data points with a high similarity or strong correlation, and finally, the data points with a high degree of relevance fall into the same or adjacent leaf nodes.

The mass distribution is exploited to stretch the fringe points away from the core points for the separation of the target and the background. Then, the critical requirement for the detection output is to highlight the target and suppress the background. In the mass space generated by the tree-structured mapping, for a test pixel and the given target spectrum, the mass value of the highest-level node shared by them reflects the degree of relevance. The smaller the mass value, the smaller the volume of the highest-level shared region, which corresponds to a high similarity or strong correlation, indicating that they are more likely to belong to the same type of land cover. Accordingly, when implementing ME based on the tree structure, the mass value of the highest-level shared node can be utilized as a basis for detection output. It is worth noting that HSIs carry complex information about various types of land cover, and target detection aims to locate targets of interest in a scene at the pixel level, which requires high evaluation accuracy. However, such high-precision requirements for detection output cannot be met only using the mass

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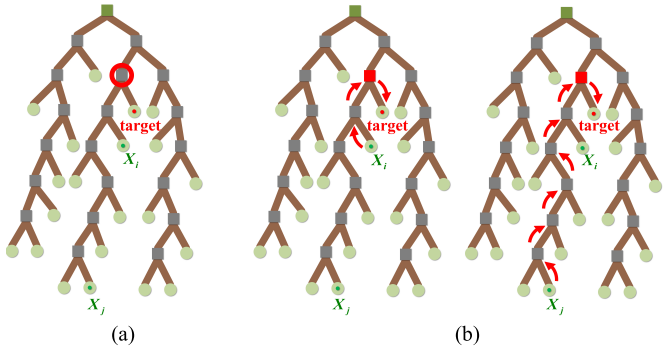


Fig. 4. Ranking measure in the tree-structured feature space. (a) Mass. (b) Topological mass.

measure for the shared node. For example, assume that $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_N]$, $\mathbf{X} \in \mathbb{R}^{L\times N}$ is an HSI dataset, and $L$ and $N$ are the numbers of bands and pixels, respectively. $\mathbf{x}_i$, $\mathbf{x}_j \in \mathbb{R}^{L\times 1}$ are two pixels in $\mathbf{X}$, and $\mathbf{target} \in \mathbb{R}^{L\times 1}$ indicates the given target spectrum. When $\mathbf{x}_i$, $\mathbf{x}_j$, and $\mathbf{target}$ are put into the root node, these three samples fall into different leaf nodes on the binary tree, as shown in Fig. 4(a). It can be observed that, in the space generated by the tree-structured mapping, the node marked by a red circle is the highest-level region shared by $\mathbf{x}_i$, $\mathbf{x}_j$, and $\mathbf{target}$. Moreover, $\text{mass}(\mathbf{x}_i, \mathbf{target})$ and $\text{mass}(\mathbf{x}_j, \mathbf{target})$, respectively, indicate the corresponding mass values. It is obvious that the mass values of the same nodes on the tree are equal, which means that they have the same degree of relevance with the given query. As mentioned earlier, the spectral difference in a HSI is essentially reflected in the numerical separability, which enables the binary tree to exploit such data characteristics to separate pixels from different types of land cover. It is obvious from the tree structure shown in Fig. 4(a) that the overall spectral similarity between $\mathbf{x}_i$ and $\mathbf{target}$ is higher than that of $\mathbf{x}_j$. Merely measuring the data mass of the shared node cannot guarantee that such difference in the degree of similarity between $\mathbf{x}_i$, $\mathbf{x}_j$, and the given $\mathbf{target}$ will be highlighted. Therefore, directly performing the measurement of data mass in the feature space cannot accurately quantify the spectral difference for detection output.

To overcome the aforementioned drawback of IR when using the mass measure for hyperspectral target detection, a specialized measurement for detection output is developed, called topological mass. By making full use of tree topology that reflects the graphical connection relationship between all nodes, we innovatively combine the mass measure with tree topology to take advantage of both. The calculation of topological mass is carried out by weighting the data mass with the path length between nodes on the tree. The above analysis of the example in Fig. 4(a) reveals that the spectral difference between test pixels and the given target spectrum can be intuitively shown in the tree structure. This inspires the idea that the shortest path length between leaf nodes on the tree where the samples finally fall can also provide an alternative basis for measuring the degree of relevance or similarity between data points and the given query. As shown in Fig. 4(b), the paths from $\mathbf{x}_i$ and $\mathbf{x}_j$ to $\mathbf{target}$ are marked with red arrows, and the path lengths are represented by $pl(\mathbf{x}_i, \mathbf{target})$ and $pl(\mathbf{x}_j, \mathbf{target})$, respectively. According to the minimum number of branches passing between two leaf nodes in Fig. 4(b), we have

$$\begin{aligned} pl(\mathbf{x}_i, \mathbf{target}) &= 3 \\ pl\left(\mathbf{x}_j, \mathbf{target}\right) &= 7. \end{aligned} \tag{1}$$

In this way, although $\mathbf{x}_i$ and $\mathbf{x}_j$ share the same highest-level region with $\mathbf{target}$, indicating that the mass measure results are the same, the difference in the degree of similarity between these two pixels and the target spectrum can still be highlighted by the path length. Based on this, the drawback resulting from only using the mass measure for the detection output can be overcome by weighting with the path length of the tree structure

$$\begin{aligned} \text{mass}_{\text{topology}}(\mathbf{x}_i, \mathbf{target}) &= \text{mass}(\mathbf{x}_i, \mathbf{target}) \cdot pl(\mathbf{x}_i, \mathbf{target}) \\ \text{mass}_{\text{topology}}\left(\mathbf{x}_j, \mathbf{target}\right) &= \text{mass}\left(\mathbf{x}_j, \mathbf{target}\right) \cdot pl\left(\mathbf{x}_j, \mathbf{target}\right). \end{aligned} \tag{2}$$

The obtained measurement in (2), topological mass, as a combination of mass measure and tree topology, takes the advantage of both. For a test pixel $\mathbf{x} \in \mathbb{R}^{L\times 1}$, the smaller the topological mass, $\text{mass}_{\text{topology}}(\mathbf{x}, \mathbf{target})$, between it and the given target spectrum, the higher the spectral similarity, and the greater the likelihood of $\mathbf{x}$ being judged as the target. Such a specialized measurement for hyperspectral target detection can effectively quantify the spectral difference, achieve highlighting the target while suppressing the background, and meet the high-precision requirements.

### C. Information Retrieval Based on an Ensemble for Hyperspectral Target Detection

When generating a feature space, the subset used to estimate a mass distribution is constructed by randomly selecting pixels from an HSI. Since no prior information is utilized at this stage, if the selected subset does not contain any target pixels, it is difficult to guarantee the effective extraction of features for the target and background. According to the concept of ensemble learning [80], in order to solve this difficulty and further improve the robustness of the proposed EIRME algorithm, the IR is performed in the feature space containing multiple mass distributions to implement parallel measurements of spectral differences. For an ensemble-based IR task with ME, an ensemble of binary trees is constructed at first. The independent trees in the ensemble correspond to parallel mass distributions, which are prepared for subsequent parallel measurements of topological mass. The aforementioned two components of ME for ensemble-based IR are evolved into the following two stages.

*1) Component 1:* An ensemble for estimating multiple mass distributions in the feature space is formed. The estimation of mass distribution is based on a tree structure. In the ensemble, the construction of each binary tree needs to randomly select $\text{sub}_{\text{size}}$ pixels from the original HSI $\mathbf{X} \in \mathbb{R}^{L\times N}$ to form a subset $\mathbf{X}_{\text{sub}} \in \mathbb{R}^{L\times \text{sub}_{\text{size}}}$. In practical applications, given that the HSIs in different real scenes have different data sizes, it is more

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reasonable to set the number of pixels used to construct the subset as a percentage of the total number of pixels

$$\text{sub}_{\text{size}} = N \cdot \text{sub}_{\text{percent}} \tag{3}$$

where $\text{sub}_{\text{percent}}$ represents the percentage of the number of subsampled pixels in $\mathbf{X}_{\text{sub}}$ to the total number of pixels in $\mathbf{X}$. The selection of an attribute $\mathbf{X}_{\text{sub_dim}}$ corresponding to a certain band is random, and a value between the maximum and minimum values of $\mathbf{X}_{\text{sub_dim}}$ is randomly selected as a newly added node. Building a binary tree is a recursive process. The termination of the recursion is closely related to two variables: the current depth of the tree $\text{depth}_{\text{current}}$ and the remaining subset $\mathbf{X}_{\text{sub}}$. In each recursion, after selecting a new node added to the tree, the remaining subset can be divided into two parts, $\mathbf{X}_{\text{left}}$ and $\mathbf{X}_{\text{right}}$, according to the relationship greater or less than the node. Correspondingly, the current depth of the tree $\text{depth}_{\text{current}}$ is increased by one, the remaining subset $\mathbf{X}_{\text{sub}}$ is updated by $\mathbf{X}_{\text{left}}$ and $\mathbf{X}_{\text{right}}$, and the updated remaining subset can be used to build the left subtree and the right subtree. This process is executed recursively until the termination condition is met, that is, the current depth of the tree $\text{depth}_{\text{current}}$ reaches the depth limit $\text{depth}_{\text{limit}}$, or the remaining subset $\mathbf{X}_{\text{sub}}$ is indivisible. The meaning of $\mathbf{X}_{\text{sub}}$ being indivisible is specific: there is only one pixel in $\mathbf{X}_{\text{sub}}$. Considering that the target and the background have obvious spectral differences in some specific bands, in order to ensure that the fringe points corresponding to the target are stretched away from the core points corresponding to the background, the number of bands, $L$, is utilized to limit the depth of the tree. Based on the previous construction of one tree, the process of constructing ensemble trees through multiple subsampling is described in Algorithm 1.

*2) Component 2:* The IR is performed pixel-by-pixel in the feature space, and the results on the ensemble are averaged as the detection output. The proposed EIRME adopts the sliding dual-window detection mode. Local background information is applied to restrict the detection result for the test pixel $\mathbf{x} \in \mathbb{R}^{L\times 1}$ and the given **target** $\in \mathbb{R}^{L\times 1}$. Specifically, for a concentric dual-window, $N_b$ background pixels between the inner and outer windows are used to form $\mathbf{X}_b \in \mathbb{R}^{L\times N_b}$. Moreover, $\boldsymbol{\mu}_b \in \mathbb{R}^{L\times 1}$ is the mean vector of the local background obtained by averaging $N_b$ pixels of $\mathbf{X}_b$. The topological mass is calculated as $\text{mass}_{\text{topology}}$ between $\mathbf{x}$ and **target**, and $\mathbf{x}$ and $\boldsymbol{\mu}_b$ to obtain $\text{mass}_{\text{topology}}(\mathbf{x}, \textbf{target})$ and $\text{mass}_{\text{topology}}(\mathbf{x}, \boldsymbol{\mu}_b)$. These are compared as

$$D(\mathbf{x}) \begin{cases} \text{mass}_{\text{topology}}(\mathbf{x}, \textbf{target}) \\ \quad < \text{mass}_{\text{topology}}(\mathbf{x}, \boldsymbol{\mu}_b) \rightarrow \text{target} \\ \text{mass}_{\text{topology}}(\mathbf{x}, \textbf{target}) \\ \quad > \text{mass}_{\text{topology}}(\mathbf{x}, \boldsymbol{\mu}_b) \rightarrow \text{background}. \end{cases} \tag{4}$$

If $\text{mass}_{\text{topology}}(\mathbf{x}, \textbf{target}) < \text{mass}_{\text{topology}}(\mathbf{x}, \boldsymbol{\mu}_b)$, the spectral similarity between $\mathbf{x}$ and the given target is higher, and $\mathbf{x}$ is more likely to be judged as the target. Conversely, if the similarity between $\mathbf{x}$ and the mean vector of the local background pixels is higher, then $\mathbf{x}$ is more likely to be judged as the background. Algorithm 2 gives the specific steps of this stage.

**Algorithm 1** Ensemble ($\mathbf{X}$, $\text{sub}_{\text{percent}}$, $\text{ensemble}_{\text{size}}$)

**Inputs**: $\mathbf{X} \in \mathbb{R}^{L\times N}$-the input HSI, $\text{sub}_{\text{percent}}$-the percentage of the number of subsampled pixels to the total number of pixels, $\text{ensemble}_{\text{size}}$-the number of mass distributions.
**Output**: **ensemble**-an ensemble of binary trees.
**Initialization**: $\text{depth}_{\text{limit}} = L$, $\text{depth}_{\text{current}} = 0$.
1: determine the subsampling size by the percentage, $\text{sub}_{\text{size}} = N \cdot \text{sub}_{\text{percent}}$;
2: **for** $k = 1 : \text{ensemble}_{zise}$ **do**
3: randomly select $\text{sub}_{\text{size}}$ pixels to form a subset $\mathbf{X}_{\text{sub}} \in \mathbb{R}^{L\times sub_{\text{size}}}$;
4: randomly shuffle integers from 1 to $L$ to get $\mathbf{dim}_{\text{use}}$, which denotes the band numbers corresponding to sequential layers of tree;
---
5: **if** $\text{depth}_{\text{current}} < \text{depth}_{\text{limit}}$ and $\mathbf{X}_{\text{sub}}$ is divisible **then**
6: $\text{dim}_{\text{current}} = \mathbf{dim}_{\text{use}}(\text{depth}_{\text{current}} + 1)$, $\mathbf{X}_{\text{sub_dim}} = \mathbf{X}_{\text{sub}}\ (\text{dim}_{\text{current}},:)$;
7: randomly select a segmentation point $\text{value}_{\text{node}}$ as the node of the tree between the maximum and minimum values of $\mathbf{X}_{\text{sub_dim}}$, tree.node $= \text{value}_{\text{node}}$;
8: $\mathbf{X}_{\text{left}} = \mathbf{X}_{\text{sub}}(:, \text{find}(\mathbf{X}_{\text{sub_dim}} <= \text{value}_{\text{node}}))$;
9: $\mathbf{X}_{\text{right}} = \mathbf{X}_{\text{sub}}(:, \text{find}(\mathbf{X}_{\text{sub_dim}} > \text{value}_{\text{node}}))$;
10: $\text{depth}_{\text{current}} = \text{depth}_{\text{current}} + 1$;
11: tree.left_node $\leftarrow \mathbf{X}_{\text{sub}} = \mathbf{X}_{\text{left}}$, re-execute from step 5;
12: tree.right_node $\leftarrow \mathbf{X}_{\text{sub}} = \mathbf{X}_{\text{right}}$, re-execute from step 5;
13: **end if (step 5)**
---
14: **return** tree
15: **ensemble** = **ensemble** $\cup$ tree;
16: **end for (step 2)**
17: **return ensemble**

Accordingly, the execution flow of the proposed EIRME can be summarized as follows. First, an ensemble of binary trees is built through multiple subsampling. Second, the test pixel is selected by sliding the dual-window, and the local background information is extracted. Third, for the test pixel, the given target and the mean vector of the local background pixels and their topological masses are calculated, respectively. These values are then compared to obtain the detection result on one tree. Finally, the results on all trees in the ensemble are averaged as the detection output.

The input of the first stage of the proposed method includes two important parameters, $\text{sub}_{\text{percent}}$ and $\text{ensemble}_{\text{size}}$. The percentage for subsampling, $\text{sub}_{\text{percent}}$, determines the size of the subset required to construct a single tree structure for estimating each mass distribution. The ensemble size, $\text{ensemble}_{\text{size}}$, determines the number of mass distributions in the feature space. The influence of the setting of these two parameters on the detection results will be discussed in Section III-A. Since EIRME adopts the local detection mode, the sizes of the outer and inner windows are set according to

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**Algorithm 2** EIRME ($\mathbf{X}$, **target**, **ensemble**, $\text{win}_{\text{outer}}$, $\text{win}_{\text{inner}}$)

**Inputs**: $\mathbf{X} \in \mathbb{R}^{L\times N}$-the input HSI, **target** $\in \mathbb{R}^{L\times 1}$-the given target spectrum, **ensemble**-the ensemble of binary trees, $\text{win}_{\text{outer}}$-the size of the outer window, $\text{win}_{\text{inner}}$-the size of the inner window.

**Output**: **result** $\in \mathbb{R}^{1\times N}$-detection results.

**Initialization:** set the size of $\text{win}_{\text{outer}}$ and $\text{win}_{\text{inner}}$ according to the spatial size of the target of interest.

1: **for** $i = 1 : N$ **do**
2: $\mathbf{x}_i \in \mathbb{R}^{L\times 1} \leftarrow i$-th test pixel of $\mathbf{X}$;
3: $\mathbf{X}_b \in \mathbb{R}^{L\times N_b} \leftarrow$ local background pixels selected by the dual-window centered on $\mathbf{x}_i$;
4: average $N_b$ pixels of $\mathbf{X}_b$ to get the mean vector $\boldsymbol{\mu}_b$.
5: $result_{i_sum} = 0$;
6: **for** $k = 1 : \text{ensemble}_{\text{size}}$ **do**
7: $tree_k \leftarrow k$-th tree of **ensemble**;
8: find the highest-level region shared by $\mathbf{x}_i$ and **target** on $tree_k$, that is, the deepest shared node, then get the corresponding mass value mass($\mathbf{x}_i$, **target**);
9: record the path from $\mathbf{x}_i$ to **target** on $tree_k$, that is, the branches that must pass between the leaf nodes that they fall into, and then get the path length $pl(\mathbf{x}_i, \textbf{target})$;
10: calculate the topological mass between $\mathbf{x}_i$ and **target**, $\text{mass}_{\text{topology}}(\mathbf{x}_i, \textbf{target}) = \text{mass}(\mathbf{x}_i, \textbf{target}) \cdot pl(\mathbf{x}_i, \textbf{target})$;
11: same as above, calculate the topological mass between $\mathbf{x}_i$ and $\boldsymbol{\mu}_b$;
$\text{mass}_{\text{topology}}(\mathbf{x}_i, \boldsymbol{\mu}_b) = \text{mass}(\mathbf{x}_i, \boldsymbol{\mu}_b) \cdot pl(\mathbf{x}_i, \boldsymbol{\mu}_b)$;

---

12: **if** $\text{mass}_{\text{topology}}(\mathbf{x}_i, \textbf{target}) = 0$ **then**
13: $result_k = mass_{\text{topology}}(\mathbf{x}_i, \boldsymbol{\mu}_b)$;
14: **else**
15: $result_k = \frac{\text{mass}_{\text{topology}}(\mathbf{x}_i, \boldsymbol{\mu}_b)}{\text{mass}_{\text{topology}}(\mathbf{x}_i, \textbf{target})}$;
16: **end if (step 12)**

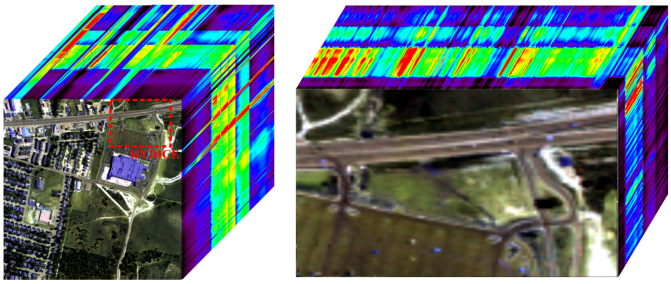
---

17: $result_{i_sum} = result_{i_sum} + result_k$;
18: **end for (step 6)**
19: $result_{i_avg} = \frac{result_{i_sum}}{tree_{num}}$;
20: $result_i = result_{i_avg}$;
21: **result** = **result** $\cup\, result_i$;
22: **end for (step 1)**
23: **return result**

the spatial size of the target of interest in different scenes for practical applications.

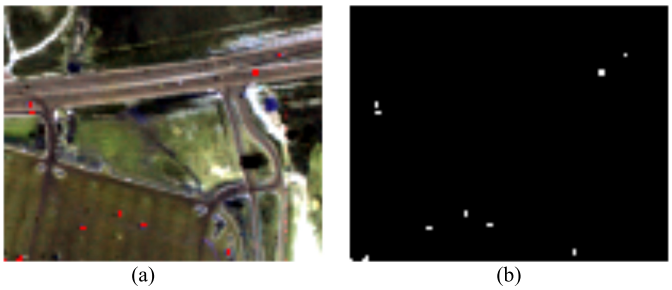
## III. Experimental Results and Analysis

Experiments are carried out on publicly available real datasets, and the experimental results are utilized to analyze the performance of the proposed method. Sections III-A and III-B introduce the benchmark experimental datasets and describe acknowledged evaluation indicators, respectively. Section III-C gives the detection results of the proposed algorithm under a variety of important parameter settings, analyzes the influence of the parameters, and finds the optimal solution. In addition, the effectiveness



(a) (b)

Fig. 5. 3-D data cube of the HYDICE. (a) Full image corresponding to the entire scene. (b) Cropped subimage for experiments.



(a) (b)

Fig. 6. 2-D display of the HYDICE. (a) True-color image. (b) Ground-truth map.

of the combination of mass measure and tree topology is demonstrated, verifying the advantages of the specialized measurement that we developed for detection output. Section III-D compares the most widely used and advanced hyperspectral target detection algorithms with EIRME to exhibit the superiority of the proposed method. The experiments in this article are run in MATLAB R2014a on a Windows 7 computer with an Intel Core i7-2600 CPU @ 3.40 GHz and 12.0-GB RAM.

### A. Experimental Datasets

Three real, publicly available HSI datasets for target detection are used in the experiments conducted for this article.

*1) HYDICE Dataset:* The first dataset was captured by the Hyperspectral Digital Imagery Collection Experiment (HYDICE) sensor. The original image covers an area in Michigan, USA, and contains 210 bands ranging from visible to near-infrared, with a spatial resolution of approximately 2 m. In total, 175 bands are preserved after removing unneeded bands, such as the low SNR and water absorption bands. As shown in Fig. 5, the full image is composed of 307 × 307 pixels and covers an urban scene. A subimage containing 80 × 100 pixels is cropped out and used in the experiments [81]. Among them, 21 pixels of vehicles are regarded as objects of interest to be detected. All targets are distributed in ten different locations, and the number of pixels occupied by the target at each location varies from 1 to 4. The true-color image in Fig. 6(a) illustrates the locations of the targets, which are marked in red. The corresponding ground-truth map is given in Fig. 6(b).

*2) AVIRIS-1 Dataset:* The second dataset was captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS). The full image covering an urban scene in San Diego contains

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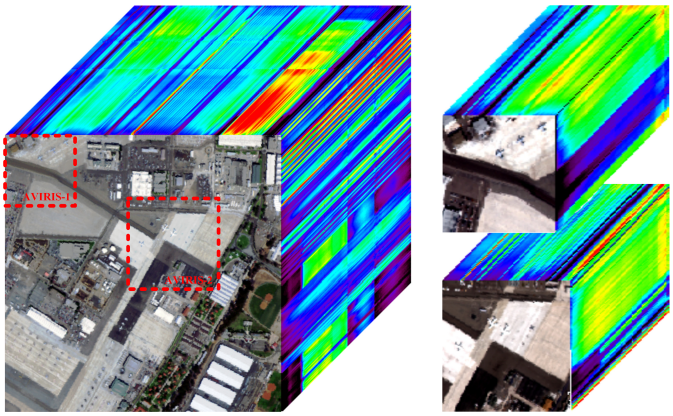
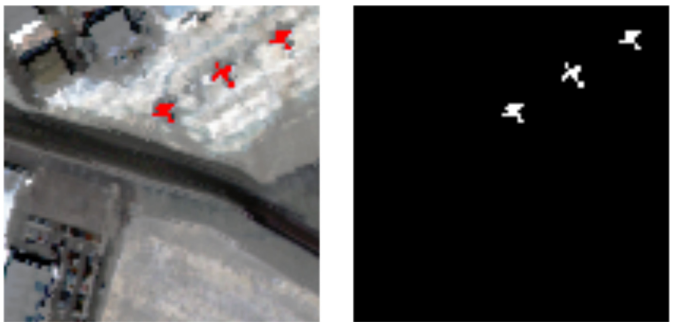



(a) (b)

Fig. 7. 3-D data cube of the AVIRIS. (a) Full image corresponding to the entire scene. (b) Cropped subimages for experiments.



(a) (b)

Fig. 8. 2-D display of the AVIRIS-1. (a) True-color image. (b) Ground-truth map.

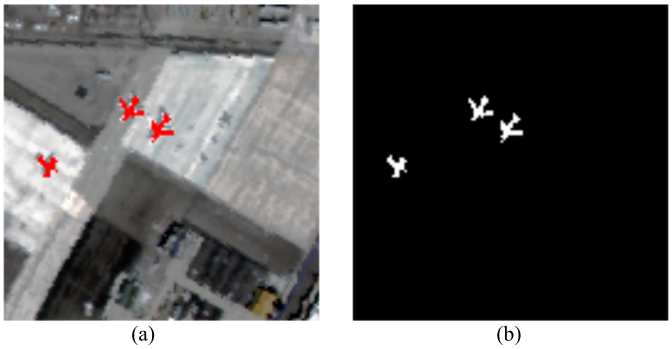
400 × 400 pixels, and the spatial resolution is 3.5 m per pixel. As shown in Fig. 7, AVIRIS-1 is a subimage containing 100 × 100 pixels selected from the entire scene. The original image contains 224 bands spanning the wavelength range of 370–2510 nm. After removing the water absorption and low SNR bands, 189 bands are reserved for the experiments [43]. In the scene shown in Fig. 8(a), three airplanes occupying 22, 22, and 20 pixels are regarded as targets of interest, and Fig. 8(b) illustrates the ground truth.

*3) AVIRIS-2 Dataset:* The third dataset was also captured by the AVIRIS. This is another subimage containing 128 × 128 pixels cut from the full AVIRIS image shown in Fig. 7. After filtering out bands that seriously interfere with the detection results, 189 bands are retained in the 3-D data cube used in our experiments [37]. There are three airplanes to be detected, which occupies 34, 42, and 44 pixels, respectively. The distribution of land cover and the locations of targets in the scene of AVIRIS-2 are shown in Fig. 9.

### *B. Evaluation Indicators*

The following part introduces several acknowledged evaluation indicators for hyperspectral target detection, which are necessary to implement a reasonable analysis and fair comparison of the experimental results in Sections III-C and III-D.

*1) Receiver Operating Characteristic (ROC) Curve:* The ROC curve illustrates the corresponding relationship between



(a) (b)

Fig. 9. 2-D display of the AVIRIS-2. (a) True-color image. (b) Ground-truth map.

multiple sets of probabilities of detection (PDs) and FARs [82]. For the detection results of an HSI, multiple sets of PDs and FARs are obtained by setting multiple sets of thresholds. The FAR refers to the probability of false alarms about the existence of the target. Based on the ground-truth map, and under a certain threshold, FAR is obtained by calculating the ratio of the number of background pixels that are misjudged as the target to the total number of background pixels

$$\mathrm{FAR} = \frac{N_{\mathrm{FA}}}{N_B} \tag{5}$$

where $N_{\mathrm{FA}}$ represents the number of pixels at background locations with detection results greater than the threshold and $N_B$ is the number of pixels at the background locations. The probability of detection refers to the probability of correctly judging the existence of the target based on the detection results under a certain threshold. The targets in the ground truth are counted in units of pixels, and PD is the proportion of the detected part in the total; we have

$$\mathrm{PD} = \frac{N_D}{N_T}. \tag{6}$$

According to the ground truth, $N_D$ is the number of target pixels correctly detected, and $N_T$ refers to the total number of targets. The performance of the detection algorithm is expected to have a lower FAR and a higher PD. With FAR as the abscissa and PD as the ordinate, the more the ROC curve goes to the upper left, the better the performance.

*2) Area Under the Curve (AUC):* The AUC indicates the area under the ROC curve [44]. The ROC curve shows the relationship between PD and FAR in an intuitive form. The more the overall trend of the curve goes to the upper left, the better the detection performance. However, when the ROC curves corresponding to different detection algorithms cross each other or are relatively close, it is impossible to accurately distinguish the differences in detection performance. In this case, AUC provides a quantitative indicator by integrating the ROC. The more the ROC curve goes to the upper left, the larger the AUC, and the better the detection performance. This enables the performance of detection algorithms to be evaluated more accurately and reasonably.

*3) Separability Map:* The separability map visualizes the separability of the detection results at the target and background locations [83]. The detection results for an HSI can

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TABLE I
AUC VALUES FOR THE DETECTION OUTPUT BASED ON MASS UNDER DIFFERENT PARAMETER COMBINATIONS ON THE HYDICE

| | $ensemble_{size}$ | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| $sub_{percent}$ | **10** | **20** | **30** | **40** | **50** | **60** | **70** | **80** | **90** | **100** |
| **0.1%** | 0.4913 | 0.7569 | 0.5548 | 0.6001 | 0.6357 | 0.7871 | 0.7039 | 0.5296 | 0.6135 | 0.5120 |
| **0.2%** | 0.6740 | 0.7463 | 0.6542 | 0.5569 | 0.5650 | 0.5505 | 0.5649 | 0.7875 | 0.7772 | 0.5364 |
| **0.3%** | 0.5993 | 0.6211 | 0.4142 | 0.6190 | 0.8030 | 0.7193 | 0.6452 | **0.8207** | 0.4918 | 0.7474 |
| **0.4%** | 0.3613 | 0.5176 | 0.6304 | 0.6417 | 0.4914 | 0.7046 | 0.6730 | 0.6595 | 0.6162 | 0.4863 |
| **0.5%** | 0.5264 | 0.6553 | 0.4913 | 0.5395 | 0.8056 | 0.5076 | 0.5547 | 0.7788 | 0.4634 | 0.5659 |
| **0.6%** | 0.4994 | 0.7896 | 0.4552 | 0.5532 | 0.4811 | 0.6953 | 0.5363 | 0.5409 | 0.3977 | 0.5037 |
| **0.7%** | 0.4772 | 0.5853 | 0.7337 | 0.4166 | 0.6667 | 0.7315 | 0.5159 | 0.5815 | 0.3825 | 0.5158 |
| **0.8%** | 0.7823 | 0.4041 | 0.6928 | 0.3593 | 0.3414 | 0.5273 | 0.6375 | 0.6178 | 0.5968 | 0.5551 |
| **0.9%** | 0.3698 | 0.5859 | 0.6870 | 0.4147 | 0.5907 | 0.7209 | 0.4704 | 0.5774 | 0.5425 | 0.3919 |
| **1.0%** | 0.6918 | 0.6470 | 0.4799 | 0.6434 | 0.5793 | 0.5895 | 0.3527 | 0.5548 | 0.5616 | 0.7403 |

be divided into two sets according to the locations of the target and background in the ground-truth map. Two colored boxes are used to display the statistical ranges of these two sets of values. The horizontal bar on each box represents the median, the edges represent the 25th and 75th percentiles, and the whiskers extend from the 0.5th to 99.5th percentiles. The degree of intersection between the two boxes reflects the degree of separation between the target and background. The smaller the amount of intersection, the stronger the ability of the detection algorithm to separate the target from the background.

### C. Verification of Advantages and Discussion on Parameters

As mentioned in Section III-C, there is a drawback in the direct migration of the IR method based on ME, that is, only using the mass measure to output detection results cannot specifically meet the high-precision requirements of target detection in HSIs. In the proposed method, the mass measure is combined with the tree topology to overcome this technical difficulty, and a specialized measurement, topological mass, is developed to quantify the spectral difference for detection output. On the same multiple mass distributions in the feature space, the detection results are output based on two different measurements, namely, mass and topological mass, to verify the advantages of this novel measurement and prove the superiority of the proposed EIRME over the direct method migration. In other words, these two different ranking measures for IR are used to conduct experiments on the same ensemble for comparison.

For Component 1 of the proposed EIRME, the estimation of multiple mass distributions in the feature space requires the construction of an ensemble of binary trees. As described in Algorithm 1, the input at this stage includes two crucial parameters, $\text{sub}_{\text{percent}}$, determines the number of subsampled pixels used to construct a single tree for estimating each mass distribution, and $\text{ensemble}_{\text{size}}$ determines the number of mass distributions. For Component 2, there are two parameters that need to be manually set in Algorithm 2: the sizes of the outer and inner windows, $\text{win}_{\text{outer}}$ and $\text{win}_{\text{inner}}$. The concentric dual-window is a classic detection mode, and the sizes of the two windows need to be set according to the spatial size of the target in different hyperspectral datasets. The requirement is that the inner window can just frame the target, and most of the pixels between the two windows are background pixels. Accordingly, the sizes of the outer and inner windows of the three experimental datasets are set according to the HSI information provided in Section III-A. The sizes of the outer and inner windows for HYDICE are $13 \times 13$ and $3 \times 3$. The window sizes for AVIRIS-1 are set to $15 \times 15$ and $5 \times 5$. The dual-window sizes for AVIRIS-2 are $17 \times 17$ and $7 \times 7$.

Therefore, the discussion on parameters focuses on the influence of the combination of $\text{sub}_{\text{percent}}$ and $\text{ensemble}_{\text{size}}$ on the detection performance of the proposed EIRME and attempts to find the optimal solution. In this regard, AUC is used as a quantitative indicator to evaluate the detection results on three experimental HSIs. Moreover, the ranges of these two parameters are set separately to expand the search range of the combination. It is worth noting that, if the size of the ensemble is too small, there are only a few mass distributions in the feature space. Moreover, if the utilization rate of subsampling to construct a subset for estimating each mass distribution is too low for the total pixels in an HSI, it is difficult to guarantee the effective extraction of features for the target and background. With these concerns, the setting range for $\text{sub}_{\text{percent}}$ is [0.1%, 0.2%, 0.3%, 0.4%, 0.5%, 0.6%, 0.7%, 0.8%, 0.9%, 1.0%], and the range for $\text{ensemble}_{\text{size}}$ is [10, 20, 30, 40, 50, 60, 70, 80, 90, 100].

For the HYDICE, according to the given ranges of $\text{sub}_{\text{percent}}$ and $\text{ensemble}_{\text{size}}$, there are a total of 100 parameter combinations, and each combination corresponds to the construction of an ensemble. Table I gives the AUC values for the detection results output based on the mass on 100 ensembles. For the same 100 ensembles, Table II gives the AUC values of the

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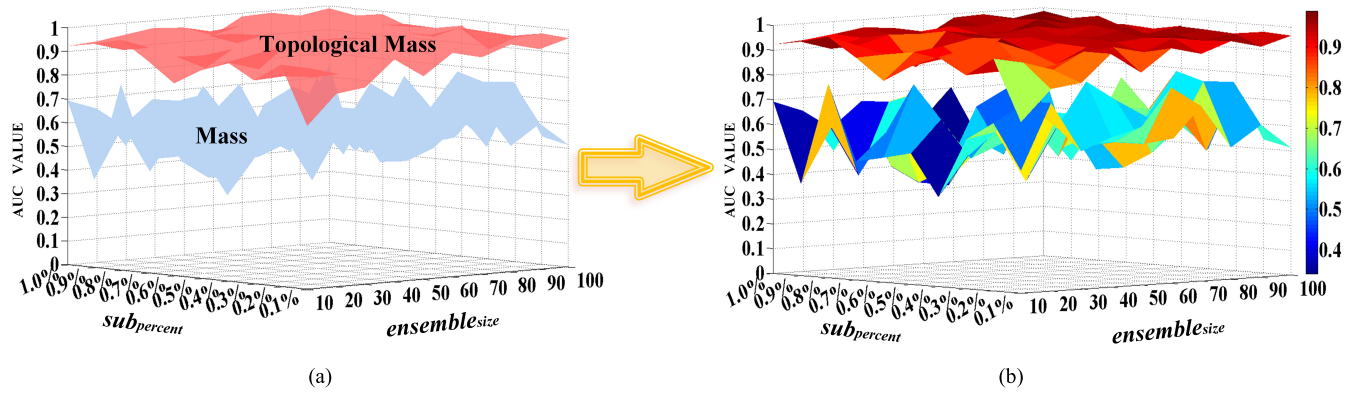
TABLE II

AUC VALUES FOR THE DETECTION OUTPUT BASED ON TOPOLOGICAL MASS UNDER DIFFERENT PARAMETER COMBINATIONS ON THE HYDICE

| | $ensemble_{size}$ | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| $sub_{percent}$ | **10** | **20** | **30** | **40** | **50** | **60** | **70** | **80** | **90** | **100** |
| **0.1%** | 0.6878 | 0.8129 | 0.8293 | 0.9368 | 0.8553 | 0.9433 | 0.9141 | 0.9576 | 0.9275 | 0.9631 |
| **0.2%** | 0.9339 | 0.9091 | 0.8733 | 0.9383 | 0.9469 | 0.9209 | 0.9687 | 0.9499 | 0.9721 | 0.9715 |
| **0.3%** | 0.8569 | 0.8633 | 0.8490 | 0.8780 | 0.9383 | 0.9521 | 0.9602 | 0.9649 | 0.9781 | 0.9430 |
| **0.4%** | 0.9219 | 0.9474 | 0.9682 | 0.9227 | 0.9459 | 0.9560 | 0.9801 | 0.9634 | 0.9831 | 0.8819 |
| **0.5%** | 0.8762 | 0.8971 | 0.8597 | 0.9649 | 0.9618 | 0.9614 | 0.9458 | 0.9622 | 0.9336 | 0.9699 |
| **0.6%** | 0.8096 | 0.9171 | 0.9330 | 0.9598 | 0.9436 | 0.9381 | 0.9340 | 0.9792 | 0.9751 | 0.9486 |
| **0.7%** | 0.8949 | 0.8997 | 0.8415 | 0.9763 | 0.9576 | 0.9542 | 0.9736 | 0.9550 | 0.9611 | 0.9424 |
| **0.8%** | 0.9704 | 0.9268 | 0.9619 | 0.9565 | 0.9140 | 0.9437 | 0.9434 | 0.9857 | 0.9605 | 0.9840 |
| **0.9%** | 0.9419 | 0.9054 | 0.9406 | 0.9477 | 0.9528 | 0.9647 | 0.9551 | 0.9765 | 0.9856 | 0.9664 |
| **1.0%** | 0.9208 | 0.9304 | 0.9486 | 0.9667 | 0.9566 | 0.9142 | 0.9826 | **0.9875** | 0.9650 | 0.9855 |




(a) (b)

Fig. 10. Illustrations of the relationship between different parameter combinations and the detection performance for two ranking measures, namely, mass and topological mass, on the HYDICE. (a) Two monochromatic surfaces. (b) Two colored surfaces.

proposed EIRME, which outputs the detection results based on topological mass. Correspondingly, Fig. 10 illustrates the contents of Tables I and II. In Fig. 10(a), the two monochromatic surfaces represent the AUC values of the detection results based on mass and topological mass, respectively. In Fig. 10(b), the two surfaces are colored according to the values, and the color bar is at the top. The AUC in Table I corresponds to mass and varies from 0.3414 to 0.8207. In Table II, AUC corresponds to topological mass and varies from 0.6878 to 0.9875, and the optimal parameter combination that maximizes the AUC value is $\text{sub}_{\text{percent}} = 1.0\%$ and $\text{ensemble}_{\text{size}} = 80$. It can be seen from Fig. 10(a) that the surface of topological mass is above the surface of mass, which illustrates that the detection performance of the proposed method using topological mass is overall ahead of that of the detector using mass. The intuitive display of experimental results demonstrates that the combination of mass measure and tree topology is effective and reveals the unique advantages of the specialized measurement that we developed. It can be observed from the two surfaces in Fig. 10(b) that the AUC value corresponding to the upper surface fluctuates more smoothly, indicating that the detection performance of the proposed method is relatively stable with the changes of $\text{sub}_{\text{percent}}$ and $\text{ensemble}_{\text{size}}$.

For the AVIRIS-1, Tables III and IV provide the respective AUC values for the detection results based on mass and topological mass under different parameter combinations. Fig. 11 shows the corresponding surfaces. The AUC value for mass varies from 0.2875 to 0.9069, and the AUC value for topological mass varies from 0.7996 to 0.9960. In Table IV, the optimal combination of parameters for the proposed method is $\text{sub}_{\text{percent}} = 0.2\%$ and $\text{ensemble}_{\text{size}} = 100$. It is observed from Fig. 11(a) that the surface of topological mass is above that of mass, indicating that using topological mass as a ranking measure for detecting output has higher accuracy than using mass. In Fig. 11(b), it is worth noting that the colored surface corresponding to topological mass is relatively smooth, and the AUC value fluctuates slightly under different parameter combinations. Within the range of parameter settings, the characteristics of the colored surface reveal that the proposed EIRME is not particularly sensitive to the parameter setting and exhibits robustness.

For the AVIRIS-2, the AUC values corresponding to the mass and topological mass are given

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TABLE III

AUC VALUES FOR THE DETECTION OUTPUT BASED ON MASS UNDER DIFFERENT PARAMETER COMBINATIONS ON THE AVIRIS-1

| | $ensemble_{size}$ | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| $sub_{percent}$ | 10 | 20 | 30 | 40 | 50 | 60 | 70 | 80 | 90 | 100 |
| **0.1%** | 0.8720 | 0.7140 | 0.8076 | 0.8859 | 0.7812 | 0.8488 | 0.8837 | 0.8473 | 0.7843 | 0.8401 |
| **0.2%** | 0.6363 | 0.4897 | 0.8207 | 0.6398 | 0.7095 | 0.8689 | 0.7783 | 0.7896 | 0.7522 | 0.8949 |
| **0.3%** | 0.7994 | 0.8583 | 0.5863 | 0.6716 | 0.6895 | 0.7249 | 0.5951 | 0.9011 | **0.9069** | 0.8134 |
| **0.4%** | 0.5888 | 0.6910 | 0.5220 | 0.5046 | 0.5826 | 0.6854 | 0.7574 | 0.5979 | 0.7180 | 0.6798 |
| **0.5%** | 0.6967 | 0.5947 | 0.5058 | 0.6178 | 0.5604 | 0.6841 | 0.5653 | 0.5482 | 0.4750 | 0.8565 |
| **0.6%** | 0.5294 | 0.6964 | 0.6163 | 0.6608 | 0.6315 | 0.5519 | 0.3301 | 0.4882 | 0.6890 | 0.7760 |
| **0.7%** | 0.5487 | 0.7042 | 0.5073 | 0.6814 | 0.4389 | 0.7287 | 0.4613 | 0.5503 | 0.2875 | 0.6250 |
| **0.8%** | 0.4537 | 0.4527 | 0.4356 | 0.5483 | 0.6085 | 0.5084 | 0.8248 | 0.5612 | 0.5430 | 0.5444 |
| **0.9%** | 0.3350 | 0.5110 | 0.5470 | 0.5157 | 0.3747 | 0.4802 | 0.8358 | 0.3588 | 0.6809 | 0.6750 |
| **1.0%** | 0.4953 | 0.4735 | 0.5741 | 0.6450 | 0.5873 | 0.4566 | 0.4853 | 0.6065 | 0.5850 | 0.5338 |

TABLE IV

AUC VALUES FOR THE DETECTION OUTPUT BASED ON TOPOLOGICAL MASS UNDER DIFFERENT PARAMETER COMBINATIONS ON THE AVIRIS-1

| | $ensemble_{size}$ | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| $sub_{percent}$ | 10 | 20 | 30 | 40 | 50 | 60 | 70 | 80 | 90 | 100 |
| **0.1%** | 0.9500 | 0.9843 | 0.9855 | 0.9946 | 0.9915 | 0.9851 | 0.9810 | 0.9909 | 0.9955 | 0.9824 |
| **0.2%** | 0.9618 | 0.9759 | 0.9877 | 0.9768 | 0.9883 | 0.9886 | 0.9912 | 0.9706 | 0.9928 | **0.9960** |
| **0.3%** | 0.9080 | 0.9857 | 0.9815 | 0.9820 | 0.9809 | 0.9921 | 0.9882 | 0.9920 | 0.9911 | 0.9909 |
| **0.4%** | 0.9846 | 0.9784 | 0.9884 | 0.9781 | 0.9842 | 0.9906 | 0.9898 | 0.9945 | 0.9874 | 0.9905 |
| **0.5%** | 0.9832 | 0.9823 | 0.9770 | 0.9790 | 0.9814 | 0.9856 | 0.9939 | 0.9876 | 0.9859 | 0.9912 |
| **0.6%** | 0.8941 | 0.9835 | 0.9739 | 0.9817 | 0.9770 | 0.9844 | 0.9765 | 0.9832 | 0.9840 | 0.9903 |
| **0.7%** | 0.7996 | 0.9784 | 0.9594 | 0.9870 | 0.9821 | 0.9913 | 0.9875 | 0.9873 | 0.9900 | 0.9908 |
| **0.8%** | 0.9623 | 0.9825 | 0.9723 | 0.9802 | 0.9798 | 0.9784 | 0.9887 | 0.9847 | 0.9901 | 0.9841 |
| **0.9%** | 0.9768 | 0.9824 | 0.9745 | 0.9830 | 0.9828 | 0.9705 | 0.9820 | 0.9829 | 0.9781 | 0.9796 |
| **1.0%** | 0.9823 | 0.9745 | 0.9840 | 0.9798 | 0.9787 | 0.9879 | 0.9840 | 0.9843 | 0.9822 | 0.9835 |

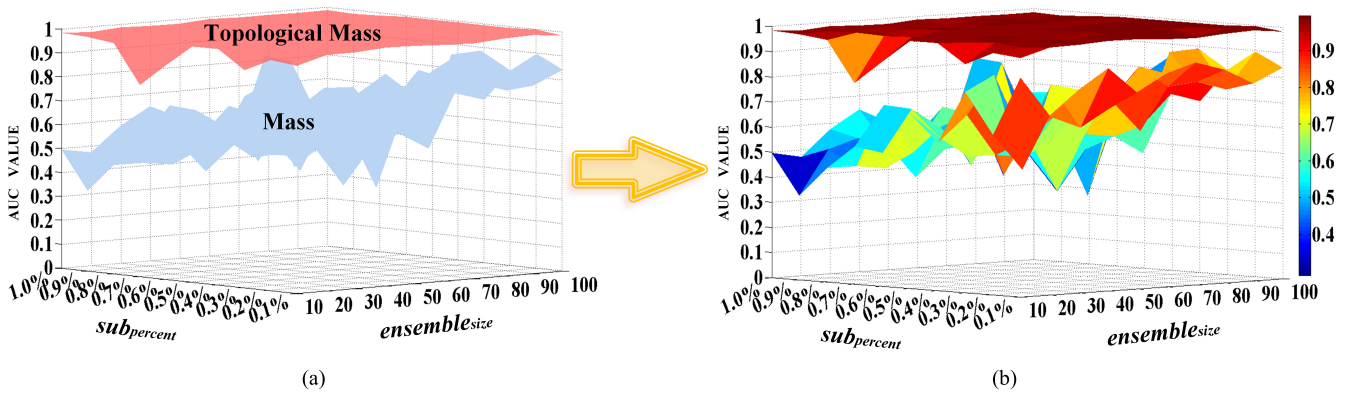


Fig. 11. Illustrations of the relationship between different parameter combinations and the detection performance for two ranking measures, namely, mass and topological mass, on the AVIRIS-1. (a) Two monochromatic surfaces. (b) Two colored surfaces.

in Tables V and VI, respectively. The AUC value for mass varies from 0.3147 to 0.7918, and the AUC value for topological mass varies from 0.7383 to 0.9366. The optimal parameter combination for the proposed method is $sub_{percent} = 0.5\%$ and $ensemble_{size} = 40$. Fig. 12 visualizes the relationship between AUC values and the combination

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TABLE V

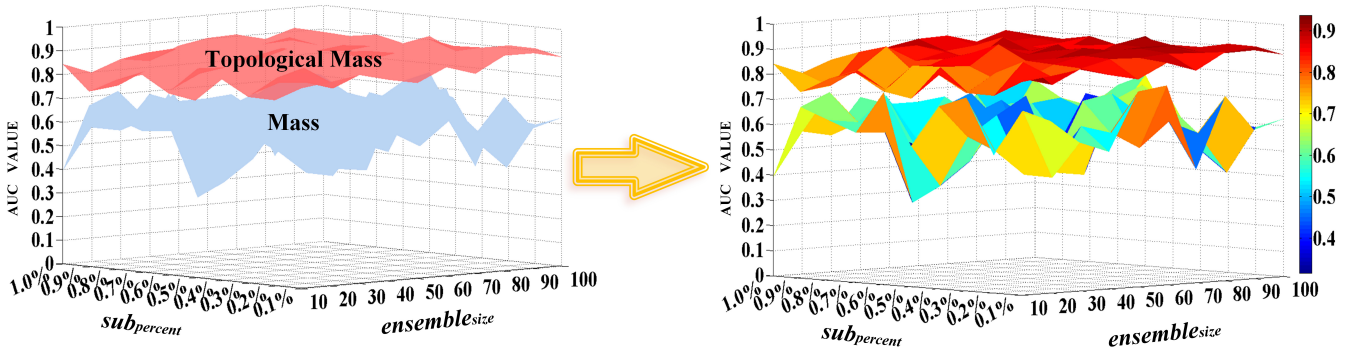
AUC VALUES FOR THE DETECTION OUTPUT BASED ON MASS UNDER DIFFERENT PARAMETER COMBINATIONS ON THE AVIRIS-2

| | $ensemble_{size}$ | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| $sub_{percent}$ | 10 | 20 | 30 | 40 | 50 | 60 | 70 | 80 | 90 | 100 |
| **0.1%** | 0.7263 | 0.6813 | 0.5885 | 0.5435 | 0.7764 | **0.7918** | 0.4507 | 0.7343 | 0.5908 | 0.6252 |
| **0.2%** | 0.5462 | 0.7145 | 0.4411 | 0.6369 | 0.6040 | 0.5345 | 0.6303 | 0.5784 | 0.4125 | 0.5848 |
| **0.3%** | 0.7657 | 0.5456 | 0.4423 | 0.4879 | 0.7221 | 0.7434 | 0.6539 | 0.6737 | 0.4966 | 0.5990 |
| **0.4%** | 0.7243 | 0.5928 | 0.6718 | 0.7417 | 0.4264 | 0.4105 | 0.7000 | 0.7524 | 0.6852 | 0.4661 |
| **0.5%** | 0.5585 | 0.3906 | 0.4786 | 0.6332 | 0.5003 | 0.5140 | 0.7183 | 0.6113 | 0.6925 | 0.5664 |
| **0.6%** | 0.7636 | 0.3147 | 0.6856 | 0.7357 | 0.4792 | 0.4613 | 0.4873 | 0.5545 | 0.6031 | 0.7593 |
| **0.7%** | 0.5940 | 0.5843 | 0.5308 | 0.5795 | 0.6328 | 0.6843 | 0.5929 | 0.5111 | 0.6162 | 0.7106 |
| **0.8%** | 0.6346 | 0.6192 | 0.7178 | 0.5408 | 0.6960 | 0.5935 | 0.6489 | 0.6467 | 0.6607 | 0.6199 |
| **0.9%** | 0.6790 | 0.7300 | 0.6000 | 0.6657 | 0.6658 | 0.6359 | 0.6118 | 0.5259 | 0.7255 | 0.5158 |
| **1.0%** | 0.3904 | 0.5642 | 0.5410 | 0.6848 | 0.5593 | 0.4066 | 0.5919 | 0.6648 | 0.7253 | 0.4526 |

TABLE VI

AUC VALUES FOR THE DETECTION OUTPUT BASED ON TOPOLOGICAL MASS UNDER DIFFERENT PARAMETER COMBINATIONS ON THE AVIRIS-2

| | $ensemble_{size}$ | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| $sub_{percent}$ | 10 | 20 | 30 | 40 | 50 | 60 | 70 | 80 | 90 | 100 |
| **0.1%** | 0.8438 | 0.8597 | 0.8788 | 0.8479 | 0.8236 | 0.8798 | 0.8417 | 0.9077 | 0.9072 | 0.8802 |
| **0.2%** | 0.7815 | 0.9084 | 0.8100 | 0.8658 | 0.8962 | 0.8914 | 0.9003 | 0.8950 | 0.8874 | 0.9056 |
| **0.3%** | 0.7838 | 0.8558 | 0.8233 | 0.8732 | 0.8991 | 0.9130 | 0.9126 | 0.9129 | 0.8848 | 0.9072 |
| **0.4%** | 0.8467 | 0.7716 | 0.8122 | 0.8803 | 0.8801 | 0.9150 | 0.8947 | 0.9023 | 0.9037 | 0.8864 |
| **0.5%** | 0.7455 | 0.8639 | 0.8573 | **0.9366** | 0.8880 | 0.8978 | 0.8544 | 0.9042 | 0.9238 | 0.8600 |
| **0.6%** | 0.7642 | 0.8602 | 0.8539 | 0.8927 | 0.9124 | 0.8741 | 0.9054 | 0.8914 | 0.8887 | 0.9119 |
| **0.7%** | 0.8313 | 0.7574 | 0.8159 | 0.8606 | 0.8511 | 0.8881 | 0.8694 | 0.8581 | 0.9104 | 0.8704 |
| **0.8%** | 0.7981 | 0.8246 | 0.9101 | 0.8463 | 0.8736 | 0.8331 | 0.9154 | 0.8717 | 0.8784 | 0.8953 |
| **0.9%** | 0.7383 | 0.7525 | 0.8817 | 0.8848 | 0.8899 | 0.8598 | 0.8748 | 0.8804 | 0.8885 | 0.8776 |
| **1.0%** | 0.8430 | 0.7967 | 0.8421 | 0.8237 | 0.8779 | 0.8998 | 0.9043 | 0.8744 | 0.9110 | 0.8833 |




(a) (b)

Fig. 12. Illustrations of the relationship between different parameter combinations and the detection performance for two ranking measures, namely, mass and topological mass, on the AVIRIS-2. (a) Two monochromatic surfaces. (b) Two colored surfaces.

of $sub_{percent}$ and $ensemble_{size}$. As shown in Fig. 12(a), the topological mass utilized by the proposed EIRME is ahead of mass on the same 100 ensembles corresponding to 100 combinations. In Fig. 12(b), the surface of topological

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TABLE VII
EVALUATION RESULTS FOR COMPARISON OF RANKING MEASURES ON EXPERIMENTAL DATASETS

| Experimental Data Set | Evaluation Results | | | | | | Optimal Parameter Setting | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Mass | | | Topological Mass | | | Mass-Based Detector | | Proposed EIRME | |
| | Minimum | Maximum | Average | Minimum | Maximum | Average | $sub_{percent}$ | $ensemble_{size}$ | $sub_{percent}$ | $ensemble_{size}$ |
| HYDICE | 0.3414 | 0.8207 | 0.5851 | 0.6878 | 0.9875 | 0.9354 | 0.3% | 80 | 1.0% | 80 |
| AVIRIS-1 | 0.2875 | 0.9069 | 0.6336 | 0.7996 | 0.9960 | 0.9798 | 0.3% | 90 | 0.2% | 100 |
| AVIRIS-2 | 0.3147 | 0.7918 | 0.6035 | 0.7383 | 0.9366 | 0.8670 | 0.1% | 60 | 0.5% | 40 |

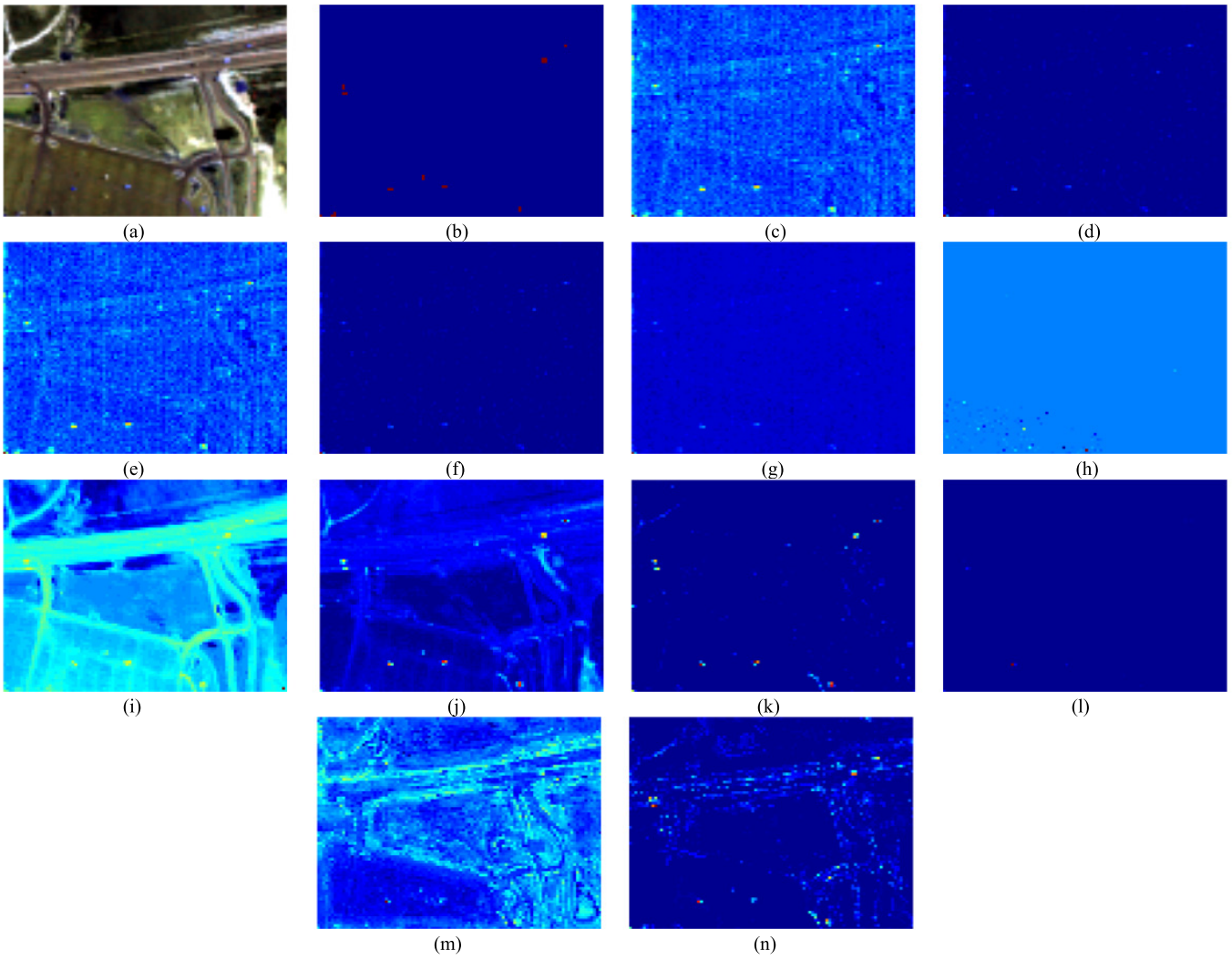


Fig. 13. Colored maps for comparison of detection algorithms on the HYDICE dataset. (a) True color. (b) Ground truth. (c) MF. (d) GLRT. (e) CEM. (f) ACE. (g) S-ACE. (h) SRD. (i) CSCR. (j) SLRMD. (k) DM-BDL. (l) TD-TSE. (m) MD. (n) EIRME.

mass fluctuates within a small interval of AUC values, which further illustrates that the proposed method is stable within the reasonable setting ranges for parameters.

In the experiments on all datasets, the detection results based on topological mass have better performance than those based on mass and proved to be more advantageous on the same ensemble. For these two different ranking measures, the evaluation results on the same ensembles are summarized in Table VII. Based on this, it can be concluded that the proposed method successfully overcomes the drawbacks of the direct migration of the IR method based on ME. The combination of mass measure and tree topology is effective, and the specialized measurement developed in this article is more suitable for target detection in HSIs. In addition, by observing Figs. 10–12, the fluctuation amplitude of the AUC colored surfaces for the proposed EIRME is different on three HSIs, and it is the smallest on the AVIRIS-1. Compared with the AVIRIS-1, the spatial sizes of targets in other experimental datasets are quite different, which results in that the effective features for the target and background cannot

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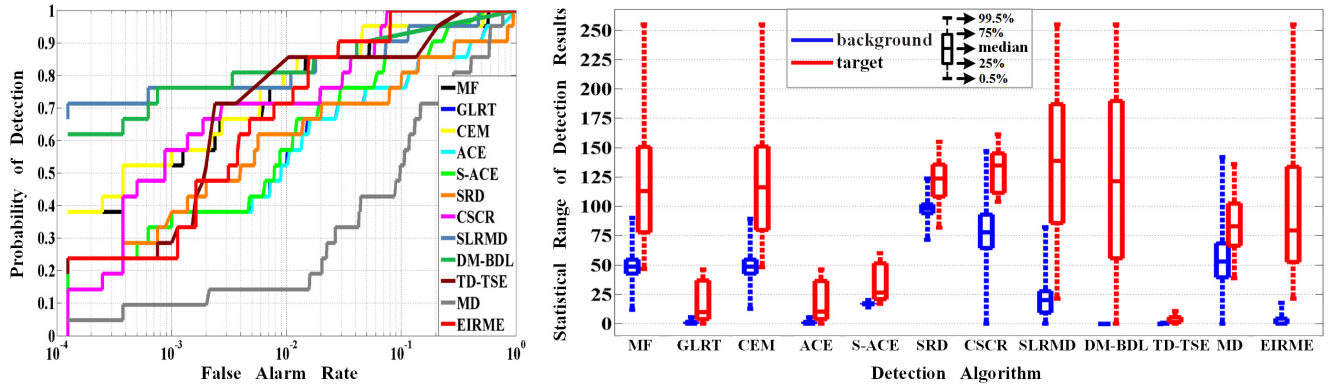


(a) (b)

Fig. 14. (a) ROC curves and (b) separability map for different detection algorithms on the HYDICE dataset.

be fully extracted when the subsampling percentage and the ensemble size are too small. Therefore, in Figs. 10 and 12, the fluctuation amplitude of the AUC colored surfaces for the proposed EIRME is relatively large. As $\text{sub}_{\text{percent}}$ and $\text{ensemble}_{\text{size}}$ increase, the AUC value becomes larger, and the surface tends to be smoother. As a conclusion, if the spatial sizes of targets of interest in an HSI are quite different, the advantages of the proposed method can be exerted by increasing the subsampling percentage and the ensemble size. Generally speaking, the overall fluctuations of three surfaces of topological mass are not particularly large, which means that, under a wide range of different parameter combinations, the AUC value for the proposed method varies within a smaller range. Such observation reveals that the detection performance of the proposed method is not particularly sensitive when given reasonable parameter setting ranges, indicating robustness. As mentioned earlier, the setting ranges for parameters $\text{sub}_{\text{percent}}$ and $\text{ensemble}_{\text{size}}$ are 0.1%∼1.0% and 10∼100, respectively. Table VII gives the optimal parameter combinations for the proposed EIRME on all experimental datasets. For different HSIs in real scenes, when $\text{sub}_{\text{percent}}$ is set between 0.2% and 1.0% and $\text{ensemble}_{\text{size}}$ is set between 40 and 100, the proposed method can achieve excellent detection performance. In practical applications, we recommend that $\text{sub}_{\text{percent}}$ and $\text{ensemble}_{\text{size}}$ to be set to 1.0% and 100 at most to achieve the desirable detection effect. The combination of $\text{sub}_{\text{percent}}$ and $\text{ensemble}_{\text{size}}$ determines how many samples are needed in the construction of an ensemble to estimate multiple mass distributions. For the HSIs used in the experiments, the optimal solution for parameter selection corresponds to a very low utilization rate of total pixels. This can be explained based on the principle of the algorithm. When generating feature spaces, it is essential to estimate the mass distribution based on the characteristics of the data itself. Corresponding to the IR task in the HSI, the data are characterized by a huge difference in the number of pixels between the target and the background, which needs to be retained in the subset that generates the feature space. In this case, subsampling with a small size is required. When the size of the subsampling is large, too many backgrounds in the subset will inevitably interfere with the process of separating the target, thus reducing their separability in the feature space. Therefore, unlike most methods based on the ensemble learning theory in ML, such as random forest and isolation forest, which require massive training samples to support [42], [84], the proposed method only needs a small-size subset for estimating each mass distribution to achieve ideal detection performance. Moreover, it can be extended to solve problems related to larger scale and higher dimensional datasets and is suitable for practical applications.

### *D. Comparison of Detection Performance*

After verifying the unique advantages of the developed specialized measurement, additional comparisons are conducted with the most widely used and advanced algorithms for hyperspectral target detection to further evaluate the competitiveness of the proposed method. These algorithms include: 1) matched filter (MF) [5]; 2) GLRT [25]; 3) CEM [37]; 4) ACE [21]; 5) signed ACE (S-ACE) [85]; 6) sparse representation-based detector (SRD) [86]; 7) combined sparse and collaborative representation-based detector (CSCR) [87]; 8) sparse and low-rank matrix decomposition (SLRMD) [88]; 9) decomposition model with background dictionary learning (DM-BDL) [89]; 10) target detection method through tree-structured encoding (TD-TSE) [90]; and 11) mass-based detector (MD) [72]. The eleventh algorithm, MD, is the detector based on the mass measure without a combination with tree topology, and the parameters of MD and the proposed EIRME are set according to the optimal scheme provided in Table VII. In order to make a fair comparison and fully demonstrate the effectiveness of the proposed EIRME, on each experimental dataset, all the aforementioned detection algorithms use the same given target spectrum as the prior information. The outputs of all algorithms are linearly stretched to the range of 0–255. Besides, for comparison algorithms with the concentric dual-window detection mode, the sizes of the outer and inner windows share the same appropriate settings as the proposed method. These preparations for the experiment are designed to allow the comparison algorithms to achieve excellent detection performance, make fair evaluations, and fully demonstrate the superiority of the proposed method.

For the HYDICE, Fig. 13 gives the detection results in the form of 2-D maps colored according to the magnitude of the value. Fig. 14(a) and (b) shows the ROC curves and

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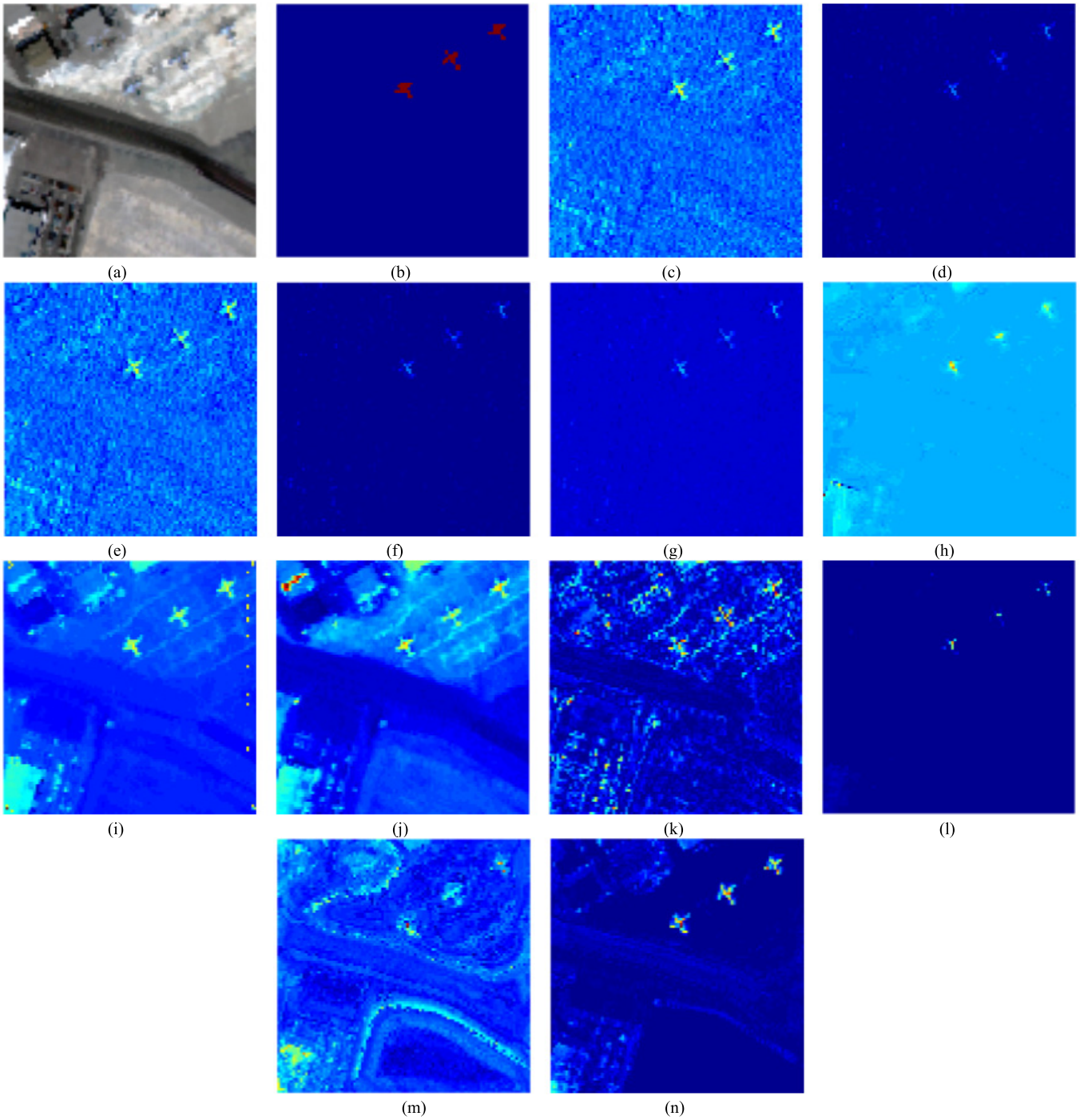


Fig. 15. Colored maps for comparison of detection algorithms on the AVIRIS-1 dataset. (a) True color. (b) Ground truth. (c) MF. (d) GLRT. (e) CEM. (f) ACE. (g) S-ACE. (h) SRD. (i) CSCR. (j) SLRMD. (k) DM-BDL. (l) TD-TSE. (m) MD. (n) EIRME.

separability map of all algorithms used in the comparison. In Fig. 13, it is obvious that the results of MD and CSCR are severely interfered with by the background, and the contours of land cover in this scene are clear. As for MF, CEM, SRD, and SLRMD, the overall maps are brighter, indicating that the detection result values are at a relatively high level, and their ability to suppress the background is limited. In contrast, GLRT, ACE, S-ACE, DM-BDL, TD-TSE, and EIRME have better effects in suppressing the background and significantly reduce the possibility of misjudging the background as the target, which corresponds to lower FARs. Among them, DM-BDL and the proposed EIRME have the best visual effect, highlighting the targets so that they could be clearly seen. In the ROC evaluation results shown in Fig. 14(a), the curves produced by GLRT, ACE, S-ACE, and SRD intersect each other and perform poorly, while the curves produced by MF, CEM, and TD-TSE are closer to the upper left corner. The detection probabilities of SLRMD and DM-BDL are quite

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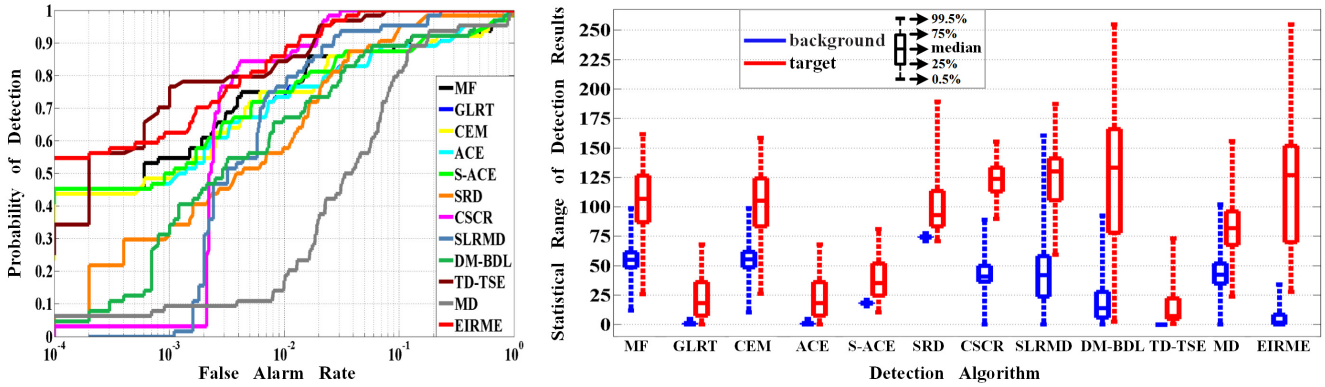


Fig. 16. (a) ROC curves and (b) separability map for different detection algorithms on the AVIRIS-1 dataset.

competitive in the low FAR region, but, with the increase in FAR, the PDs perform slightly worse. The overall ROC trend for the proposed EIRME is outstanding, and the PDs of EIRME and CSCR both reach the highest after FAR = 0.08. Compared with the above algorithms, the evaluation result for the MD developed using direct method migration is at an obvious disadvantage. Fig. 14(b) provides a comparison of the separability between the target and the background for all detectors. For MF, CEM, SRD, CSCR, SLRMD, and MD, there is a significant amount of intersection between the two colored boxes with whiskers. This illustrates that the two sets of values at the target and background locations are in a close range in the detection results, indicating a poor ability to separate the target from the background. The boxes for GLRT, ACE, S-ACE, and TD-TSE have a smaller degree of intersection and a higher degree of separation. However, an improper threshold setting is still prone to misjudgment of the target and background. For DM-BDL and EIRME, there is almost no intersection between the two boxes. An observation of the statistical range of the two sets of values reveals that the proposed method highlights the target while suppressing the background to a very low level, exhibiting prominent separation ability in comparison with other algorithms.

For the AVIRIS-1, the colored maps displaying MF, CEM, SRD, CSCR, SLRMD, and MD in Fig. 15 are brighter overall. Although the shapes of three targets are basically preserved, the detection result values at background locations are relatively large, leading to false alarms under certain thresholds. For example, the buildings in the lower left corner of the scene in these six maps are highlighted to varying degrees, which may cause the pixels at the corresponding locations to be misjudged as targets. In terms of background suppression, GLRT, ACE, S-ACE, DM-BDL, TD-TSE, and EIRME perform more prominently. However, except for the proposed EIRME, the other five algorithms pay the cost of losing most of the spatial information of the objects of interest, which means that the original shape of the targets could not be recognized well in the detection result maps. The proposed method avoids this problem and takes the lead in visual effects. Among the ROC evaluation results in Fig. 16(a), MD performs the worst, and the curves for MF, GLRT, CEM, ACE, S-ACE, SRD, CSCR, SLRMD, and DM-BDL are interlaced and also exhibit disadvantages. The curves produced by TD-TSE and EIRME are closest to the upper left, indicating the superiority of the proposed method. As for the separability map shown in Fig. 16(b), the intersection area between the two colored boxes with whiskers of MF, CEM, SLRMD, DM-BDL, and MD is much larger than that of the other algorithms, so it is difficult to separate the target from the background using their detection results. For GLRT, ACE, S-ACE, and TD-TSE, the red boxes representing the target are all in a lower position, which means that the highlighting effect for the target is not ideal. This also explains the blurring of the targets seen in the 2-D maps in Fig. 15. For the rest of SRD, CSCR, and EIRME, the intersection areas between the two boxes are very small, illustrating the strong separability between the target and background. Moreover, the detection result values of EIRME at the target locations can reach the maximum, which proves that the proposed method has the best comprehensive performance.

For the AVIRIS-2, it can be observed from Fig. 17 that, in the display of detection results for CSCR, SLRMD, and MD, the contours of land cover, such as roads and buildings in this scene, are clear, and the background interference seriously affects the judgment of the existence of the target. The overall detection result values of MF, CEM, and SRD are relatively high, and their performance in background suppression is inferior to GLRT, ACE, S-ACE, DM-BDL, TD-TSE, and EIRME. Among these six algorithms, EIRME has an absolute advantage in highlighting the target while suppressing the background. Therefore, the proposed method leads the visual comparison. Fig. 18(a) provides a comparison of all algorithms based on the ROC evaluation results; MD is still at a disadvantage. Among the twelve algorithms, the overall trends are similar for MF, GLRT, CEM, ACE, S-ACE, SLRMD, DM-BDL, TD-TSE, and EIRME. Moreover, SRD and CSCR are similar, and TD-TSE and EIRME perform best after FAR = 0.01. In the separability map shown in Fig. 18(b), the statistical ranges of the detection results at the target and background locations for MF, CEM, CSCR, SLRMD, DM-BDL, and MD overlap to a large extent, resulting in poor separability between these two types of land cover. For S-ACE and SRD, an improper threshold setting

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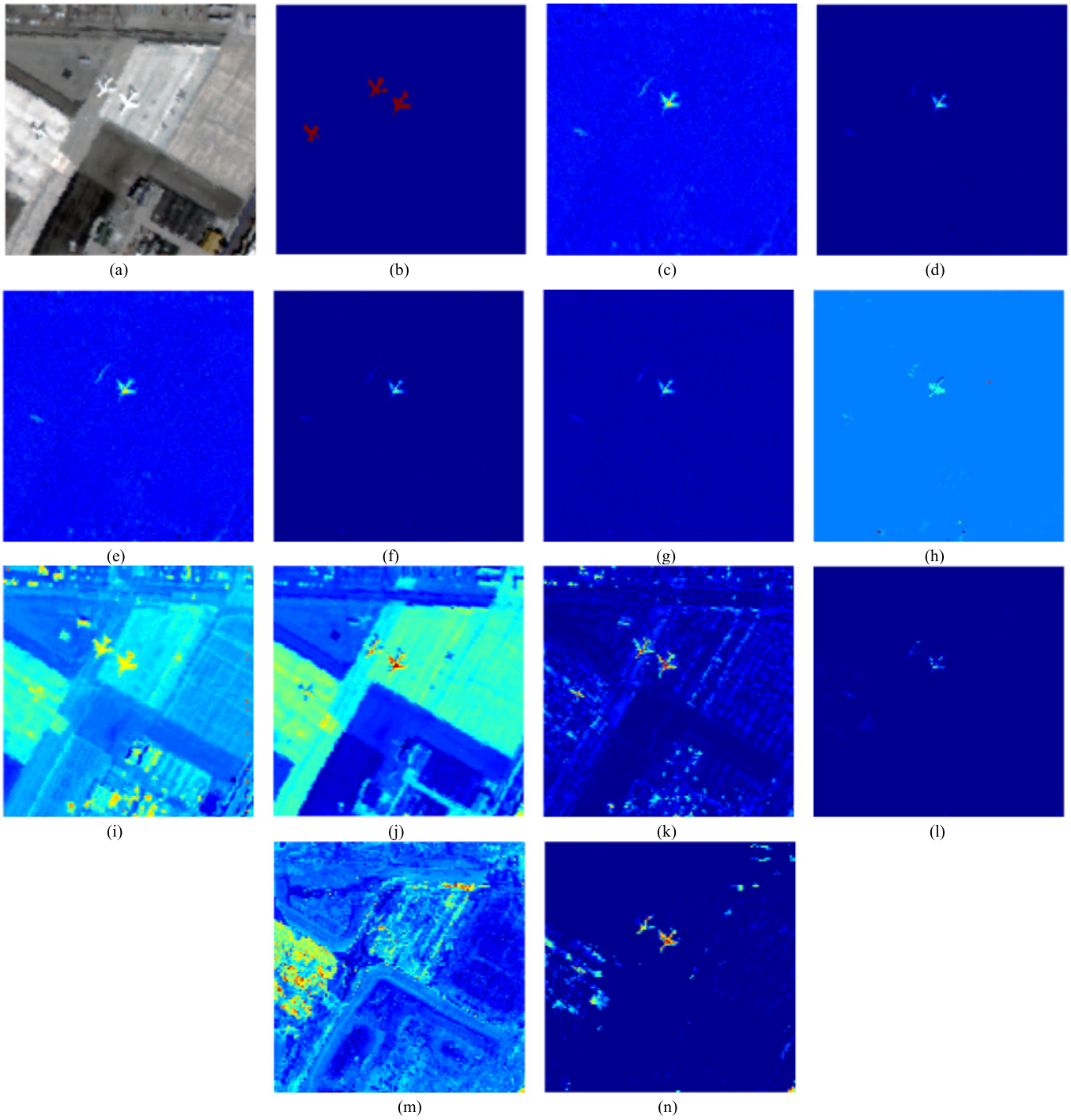


Fig. 17. Colored maps for comparison of detection algorithms on the AVIRIS-2 dataset. (a) True color. (b) Ground truth. (c) MF. (d) GLRT. (e) CEM. (f) ACE. (g) S-ACE. (h) SRD. (i) CSCR. (j) SLRMD. (k) DM-BDL. (l) TD-TSE. (m) MD. (n) EIRME.

can easily lead to misjudging the background as the target. In contrast, GLRT, ACE, TD-TSE, and EIRME suppress the background to a lower level, and the possibility of false alarms in the judgment of detection results is much smaller. Moreover, the red box with whiskers in Fig. 18(b) illustrates the outstanding ability of EIRME to highlight the target. Given a certain threshold, the proposed method has a high probability of correctly judging the existence of the target, which is superior to the other comparison algorithms.

Table VIII gives the execution time of the detection algorithms on all experimental datasets. For each algorithm, the total execution time is provided at the end of the table for comparison. Table IX summarizes the AUC evaluation results for all algorithms on the experimental HSIs to accurately distinguish the differences in detection performance through the quantitative indicator. On each dataset, the largest AUC corresponds to the algorithm with the best detection performance and is marked in bold. According to the field of

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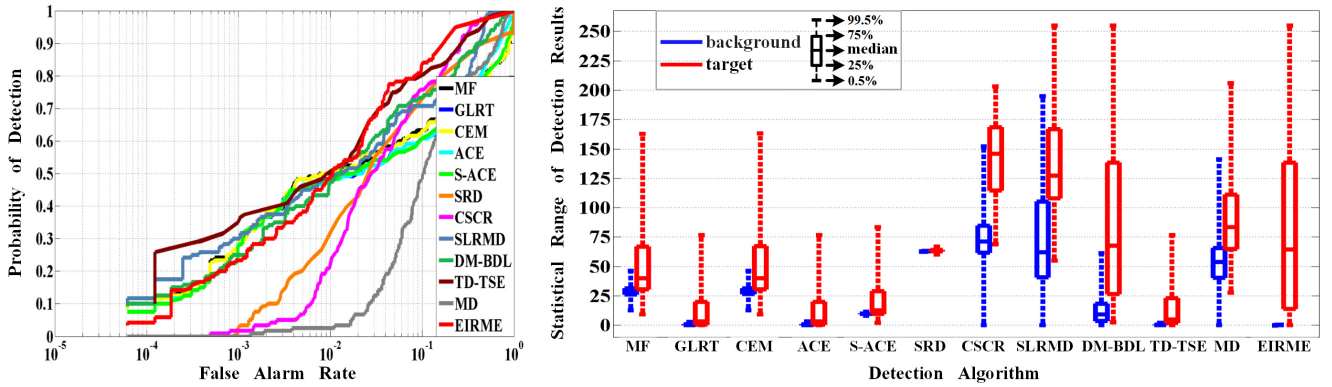


Fig. 18. (a) ROC curves and (b) separability map for different detection algorithms on the AVIRIS-2 dataset.

TABLE VIII
EXECUTION TIME FOR THE COMPARISON OF DETECTION ALGORITHMS ON EXPERIMENTAL DATASETS

| Data Set | Execution Time of the Detection Algorithms(s) | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | MF | GLRT | CEM | ACE | S-ACE | SRD | CSCR | SLRMD | DM-BDL | TD-TSE | MD | EIRME |
| HYDICE | 0.6456 | 0.8269 | 0.2868 | 0.6078 | 0.8797 | 361.7756 | 27.2601 | 33.0184 | 9.0573 | 30.3746 | 154.8898 | 277.0273 |
| AVIRIS-1 | 0.7490 | 1.1459 | 0.3182 | 0.7509 | 1.2791 | 690.405 | 41.2569 | 78.8757 | 9.5862 | 46.3367 | 207.0243 | 356.9087 |
| AVIRIS-2 | 0.9833 | 2.0926 | 0.5106 | 1.2640 | 1.7940 | 1011.20 | 95.0005 | 123.2451 | 11.7199 | 15.9416 | 208.6219 | 290.4178 |
| **Execution Time (sum)** | 2.3779 | 4.0654 | 1.1156 | 2.6227 | 3.9528 | 2063.3806 | 163.5175 | 235.1392 | 30.3634 | 92.6529 | 570.536 | 924.3538 |

TABLE IX
EVALUATION RESULTS FOR COMPARISON OF DETECTION ALGORITHMS ON EXPERIMENTAL DATASETS

| Data Set | HYDICE | AVIRIS-1 | AVIRIS-2 |
|---|---|---|---|
| **Detector** | **Evaluation Result (AUC)** | | |
| MF | 0.9649 | 0.9340 | 0.7827 |
| GLRT | 0.8965 | 0.9379 | 0.8036 |
| CEM | 0.9692 | 0.9298 | 0.7826 |
| ACE | 0.8963 | 0.9379 | 0.8034 |
| S-ACE | 0.9476 | 0.9347 | 0.7813 |
| SRD | 0.8836 | 0.9664 | 0.8533 |
| CSCR | 0.9858 | 0.9949 | 0.9146 |
| SLRMD | 0.9671 | 0.9838 | 0.8904 |
| DM-BDL | 0.9469 | 0.9374 | 0.8878 |
| TD-TSE | 0.9685 | 0.9950 | 0.9254 |
| MD | 0.8207 | 0.9069 | 0.7918 |
| **EIRME** | **0.9875** | **0.9960** | **0.9366** |

development, the 12 algorithms for comparison can be roughly divided into two categories. Traditional methods, such as MF, GLRT, CEM, ACE, and S-ACE, originate from signal processing, while SRD, CSCR, SLRMD, DM-BDL, TD-TSE, MD, and the proposed EIRME are developed based on ML. Preliminary conclusions can be drawn from the comparison results on all experimental datasets. Copious studies have shown that the stability and computational efficiency of traditional methods are undeniable. However, in most cases, the algorithms developed based on ML have certain advantages over traditional signal processing-based methods in detection effects at the cost of execution time. This phenomenon conforms to and reflects the current development trend of methodological research for hyperspectral target detection. It is worth noting that MD, as a direct migration of the IR method with mass measure, could highlight the target and suppress the background to a certain extent. However, it cannot compete with other more mature and sophisticated algorithms in terms of performance evaluation. Yet, MD has the ability to meet the essential requirements of the detector, which provides a basis for the proposed EIRME to break through technical difficulties. The AUC values for EIRME are the largest on all experimental datasets and proved to be more advantageous compared to the other algorithms. Since AUC is an evaluation indicator that comprehensively examines the probability of detection and FAR, the proposed method exhibits ideal comprehensive performance in comparison, which fully proves its effectiveness and superiority.

## IV. CONCLUSION

In this article, we proposed an ensemble-based IR method with ME for hyperspectral target detection. Focusing on the

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features of high-dimensional data itself, target detection in HSIs is treated as an IR task to break the constraints of traditional methods that need model assumptions. The target spectrum is taken as a given query and ME is exploited to solve this IR task. For estimating the mass distribution, a tree-structured mapping is utilized to generate a feature space in which the target pixels are stretched away from the background. According to the ensemble learning theory, the above IR task is executed on an ensemble of binary trees to improve the robustness of the proposed method. Directly implementing the measurement of data mass in the feature space cannot meet the high-precision requirements of target detection in HSIs. Therefore, the mass measure was innovatively combined with tree topology to surmount this technical difficulty, and a specialized measurement, topological mass, was developed to more effectively quantify the spectral difference for detection output. Comprehensive experiments were conducted on real HSI datasets. Experimental results demonstrated that the developed topological mass successfully overcomes the drawbacks of direct method migration. In addition, the proposed method only needs small-size subsets to form an ensemble to achieve high-fidelity detection, indicating high efficiency and robustness. In comparison with other classic and advanced algorithms, the proposed EIRME leads in both qualitative and quantitative evaluation and exhibits superior performance.

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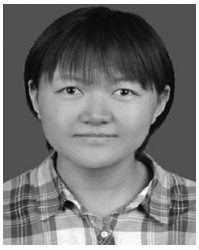

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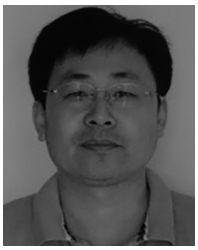

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